Cloud Security Builder Project

Securing and Monitoring Resources with AWS



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Introduction

This project involved designing, building and securing a cloud infrastructure environment using Amazon Web Services (AWS). The main objective was to develop practical experience in implementing cloud security measures, protecting sensitive data and monitoring resources against potential security threats.

Throughout the project, I worked with a range of AWS services, including Amazon VPC, Security Groups, Network Access Control Lists (NACLs), AWS Network Firewall, IAM, Amazon S3, AWS Key Management Service (KMS), CloudTrail and CloudWatch. These services were used to implement multiple layers of security, manage user permissions, restrict network traffic, encrypt data and monitor activity across the cloud environment.

Practical security testing was also carried out to verify the effectiveness of the configurations, including testing network connectivity, blocking unauthorised access, validating firewall rules and identifying compliance issues. Additionally, I implemented logging, monitoring and automated notifications to improve visibility of security events and support the ongoing protection of cloud resources.

This project allowed me to develop hands-on experience with AWS cloud security, network configuration, access management, encryption and security monitoring. It demonstrates my ability to configure and manage cloud security services while considering important industry principles such as defence in depth, least-privilege access, compliance and cost optimisation.

The project was structured around eight key objectives (R1–R8):

- R1 – Design a secure AWS network infrastructure
- R2 – Optimise Cost by evaluating the expenses associated with cloud security services
- R3 – Restrict Access through IAM policies, S3 bucket permissions and network controls
- R4 – Enforce Compliance by identifying and remediating security configuration issues
- R5 – Encrypt Data using AWS KMS to protect information at rest
- R6 – Withstand Penetration Testing by testing network security controls and identifying vulnerabilities
- R7 – Monitor and Log User Activity using CloudTrail, CloudWatch, VPC Flow Logs and S3 access logging
- R8 – Generate Alerts and Change Management Notifications to improve security awareness and incident response. Together, these objectives demonstrate the implementation of a layered cloud security solution designed to protect, monitor and maintain AWS resources.

R1 – Design

The Phase 2 network was redesigned using layered controls. The LabVPC route table was corrected to provide internet connectivity through the Internet Gateway, while the WebServer security group restricted inbound traffic to the required services. A network ACL added subnet-level filtering. The second VPC used AWS Network Firewall and dedicated route tables so that traffic to and from WebServer2 was inspected by the firewall.

Phase 2 Diagramming Activity

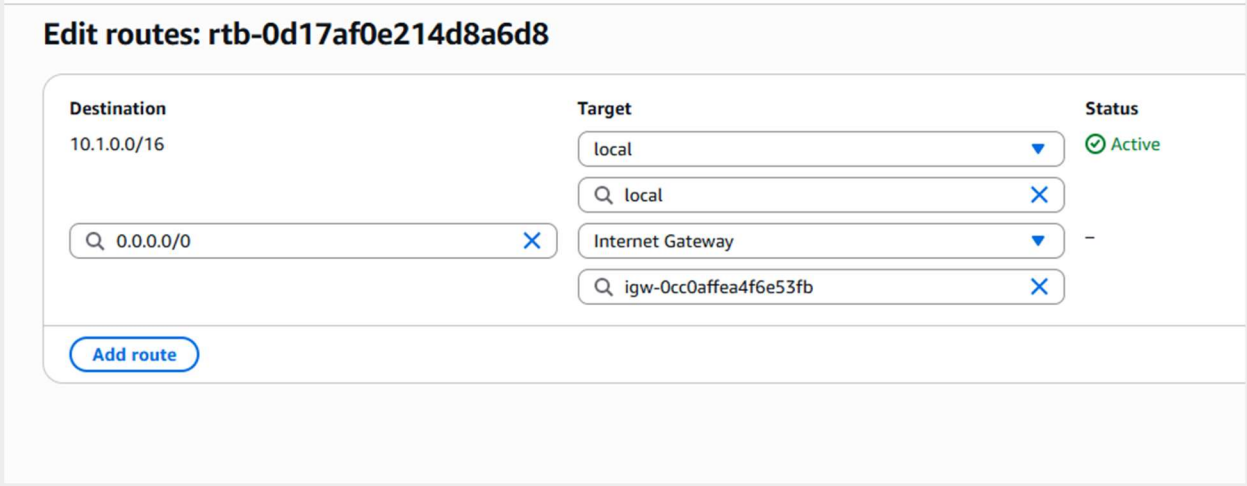
IGW-Ingress-Route-Table	
Destination	**Target**
10.1.0.0/16	local
10.1.0.0/16	vpce-0b6ba8f16a1fd56ec

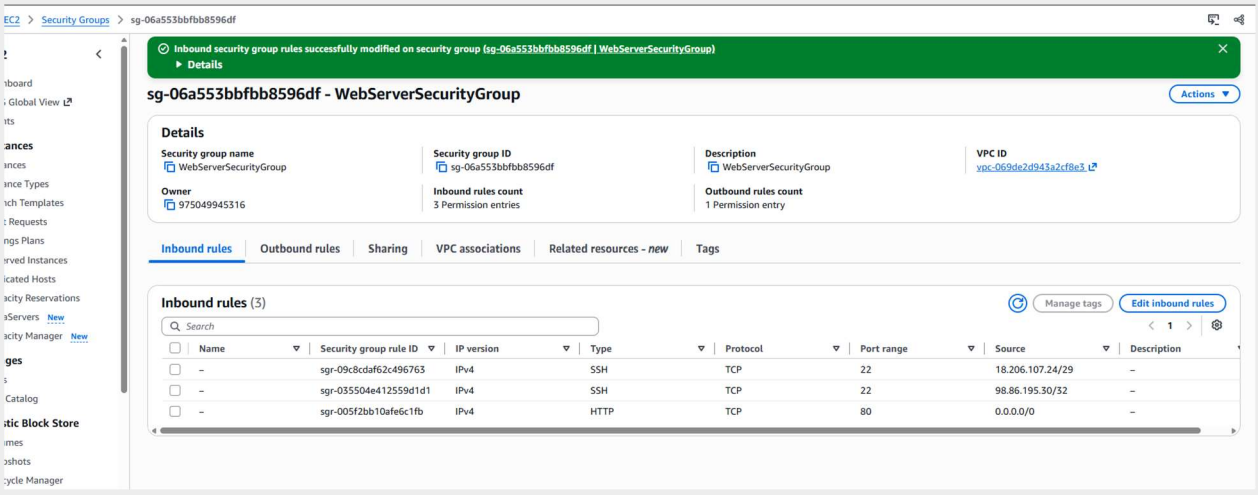
Firewall-Route-Table	
Destination	**Target**
0.0.0.0/0	igw-06f57968e90691566
10.1.0.0/16	local

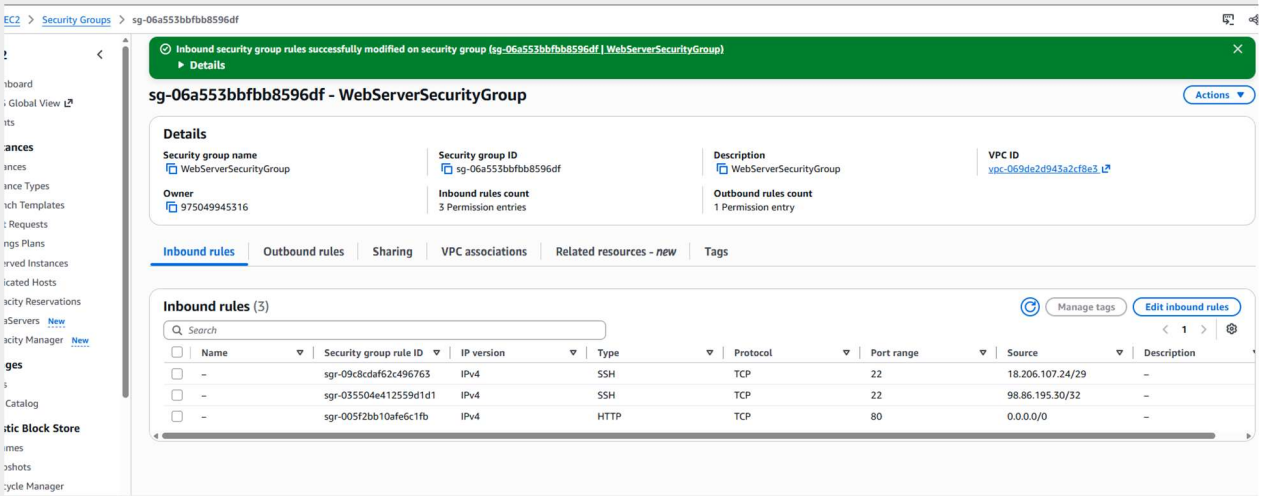
WebServer2-Route-Table	
Destination	**Target**
0.0.0.0/0	vpce-0b6ba8f16a1fd56ec
10.1.0.0/16	local

Network Firewall Rules		
Protocol	**Port**	**Action**
TCP	8080	Drop
TCP	80	Pass
TCP	22	Pass
TCP	443	Pass
ICMP	any	Pass

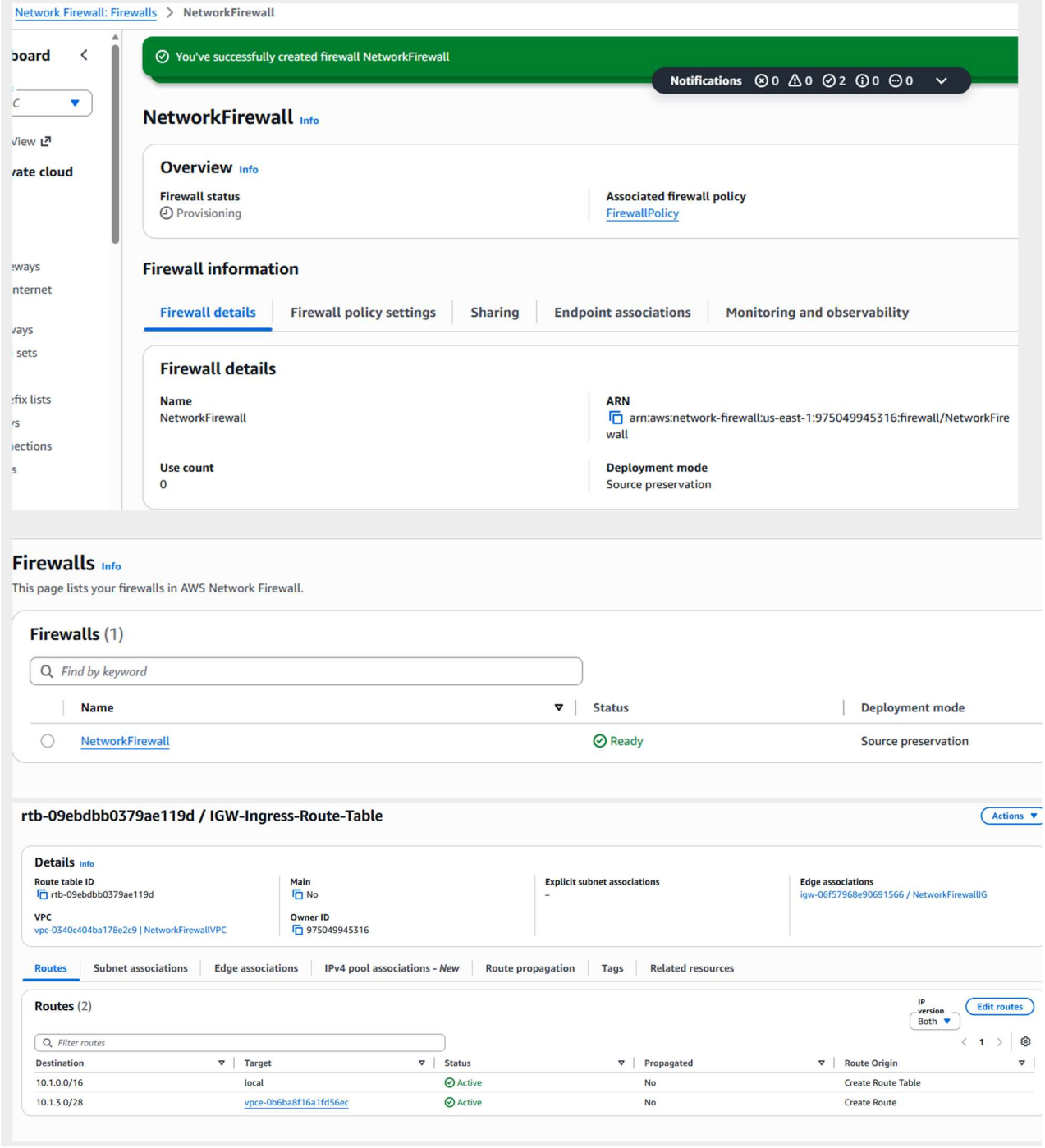
WebServerSubnet Internet Gateway route, saved routes and WebServer security-group rules.

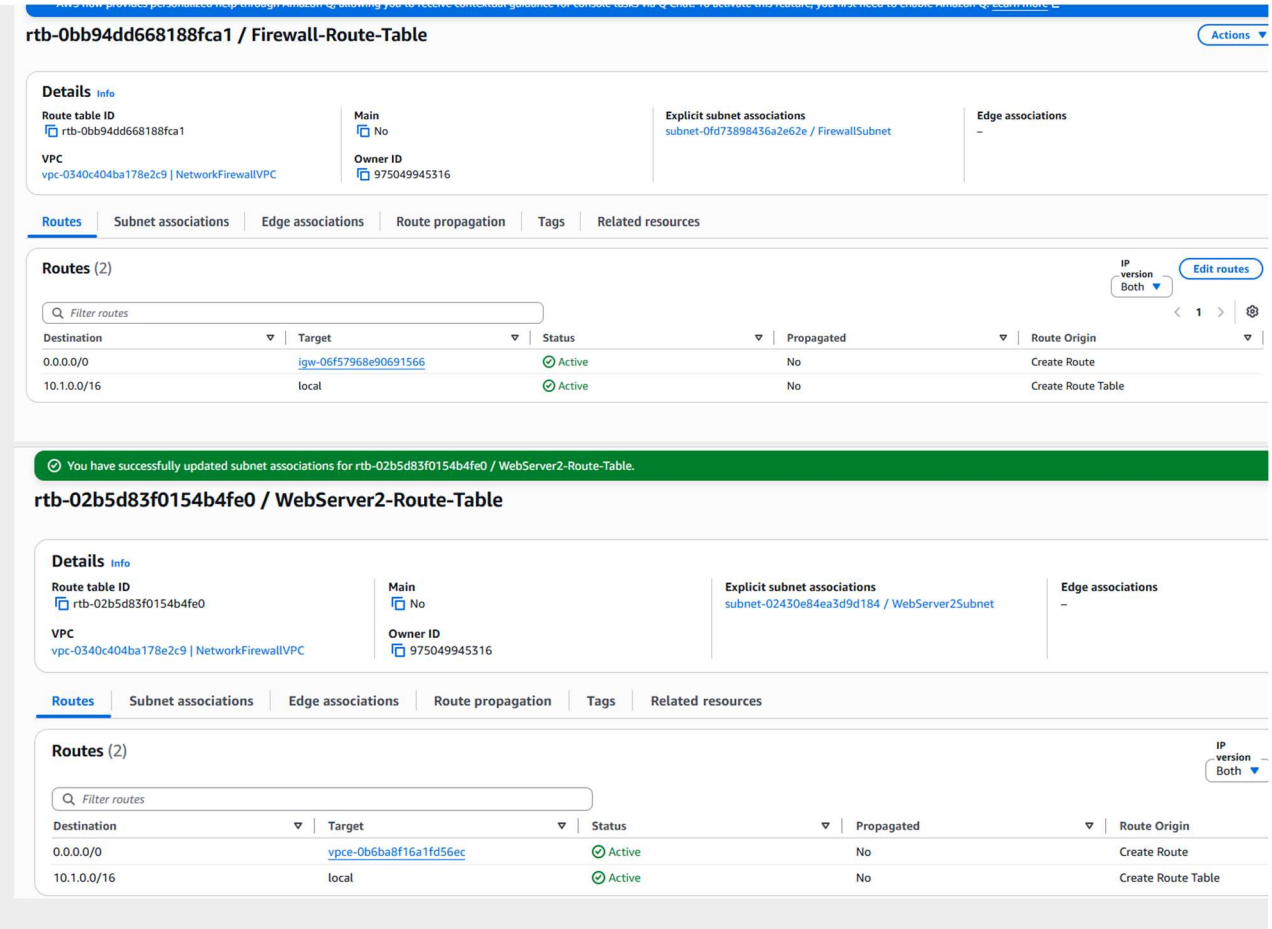




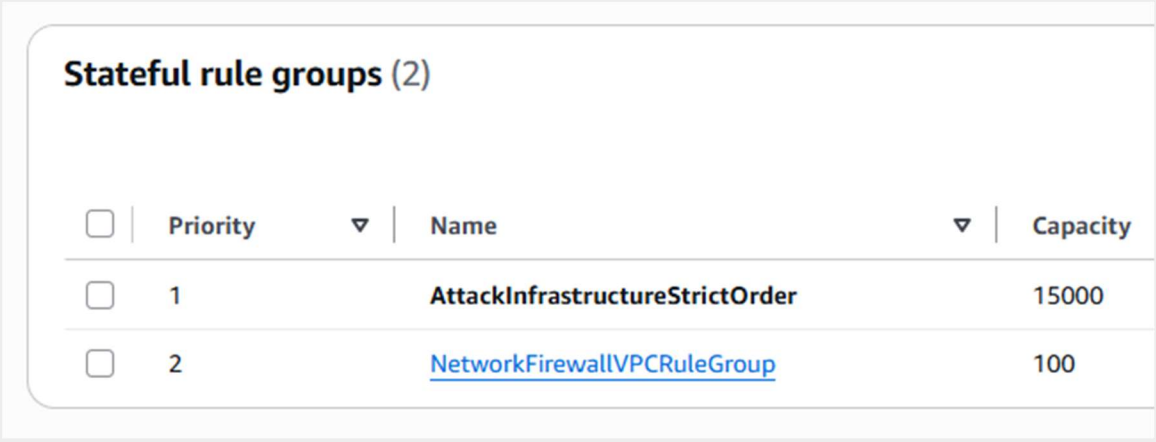


NetworkFirewall created/ready and Route Tables 1–3.





Stateful Rule Groups



Routing explanation: inbound internet traffic is associated with the Internet Gateway ingress route table and is sent to the Network Firewall endpoint before reaching WebServer2Subnet. Outbound traffic from WebServer2Subnet is also sent to the firewall endpoint before the FirewallSubnet route forwards allowed traffic to the Internet Gateway.

R2 – Optimise Cost

aws

Contact your AWS representative: Contact Sales

Export Date: 10/31/2026 **Language:** English

Estimate url

Estimate summary

Upfront cost	Monthly cost	Total 12 months cost
0.00 USD	431.03 USD	5,172.36 USD Includes upfront cost

Detailed Estimate

Name	Group	Region	Upfront cost	Monthly cost
Amazon EC2	-	US East (N. Virginia)	0.00 USD	3.07 USD
Status	-			
Description:	EC2 - WebServer2			
Config summary	Tenancy (Shared Instances), Operating system (Linux), Workload (Consistent, Number of instances: 1), Advance EC2 instance (t4g.nano), Pricing strategy (On-Demand Utilization: 100 %Utilized/Month), Enable monitoring (disabled), DT Inbound: Not selected (0 TB per month), DT Outbound: Not selected (0 TB per month), DT Intra-Region: (0 TB per month)			

Name	Group	Region	Upfront cost	Monthly cost
Amazon Virtual Private Cloud (VPC)	-	US East (N. Virginia)	0.00 USD	10.44 USD
Status	-			
Description:	VPC - NetworkFirewallVPC			
Config summary	Number of active IP addresses (1) DT Inbound: Internet (100 GB per month), DT Outbound: US East (Ohio) (1 TB per month), DT Intra-Region: (0 TB per month), Data transfer cost (10.24)			

Name	Group	Region	Upfront cost	Monthly cost
AWS Network Firewall	-	US East (N. Virginia)	0.00 USD	417.52 USD
Status	-			
Description:	-			
Config summary	Number of AWS Network Firewall endpoints (1), Number of Network Firewall secondary endpoints (1), Monthly usage per endpoint (30 days), Data processed per month (2 TB)			

Acknowledgement

AWS Pricing Calculator provides only an estimate of your AWS fees and doesn't include any taxes that might apply. Your actual fees depend on a variety of factors, including your actual usage of AWS services. Learn more

The majority of the cost comes from AWS Network Firewall, which costs $417.52 per month. The EC2 instance is much cheaper at $3.07 per month, while the VPC costs $10.44, including data transfer charges. This estimate shows that adding network security services can significantly increase the cost of cloud infrastructure. However, the firewall provides additional protection by monitoring network traffic and helping to prevent malicious or unauthorised access.

Phase 1 – S3 Bucket Access

The data bucket was protected with a bucket policy. Testing demonstrated that Paulo could access the permitted bucket and its objects, while Mary received an unauthorised/insufficient-permissions result. This proves that resource-level controls restricted access even though the users were members of the same Account Manager group.

S3 data-bucket policy

```
{
   "Version": "2012-10-17",
   "Statement": [
      {
         "Sid": "AllowApprovedUsers",
         "Effect": "Allow",
         "Principal": "*",
         "Action": "s3:*",
         "Resource": [
            "arn:aws:s3:::data-bucket-0cc0affea4f6e53fb",
            "arn:aws:s3:::data-bucket-0cc0affea4f6e53fb/*"
         ],
         "Condition": {
            "ArnEquals": {
               "aws:PrincipalArn": [
                  "arn:aws:iam::975049945316:role/voclabs",
                  "arn:aws:iam::975049945316:user/paulo",
                  "arn:aws:iam::975049945316:user/sofia"
               ]
```

Figure 1 Allow

```
{
   "Sid": "DenyUnapprovedUsers",
   "Effect": "Deny",
   "Principal": "*",
   "Action": "s3:*",
   "Resource": [
      "arn:aws:s3:::data-bucket-0cc0affea4f6e53fb",
      "arn:aws:s3:::data-bucket-0cc0affea4f6e53fb/*"
   ],
   "Condition": {
      "ArnNotEquals": {
         "aws:PrincipalArn": [
            "arn:aws:iam::975049945316:role/voclabs",
            "arn:aws:iam::975049945316:user/paulo",
            "arn:aws:iam::975049945316:user/sofia"
         ]
```

Figure 2 Deny

Paulo successful access followed by Mary unauthorised access.

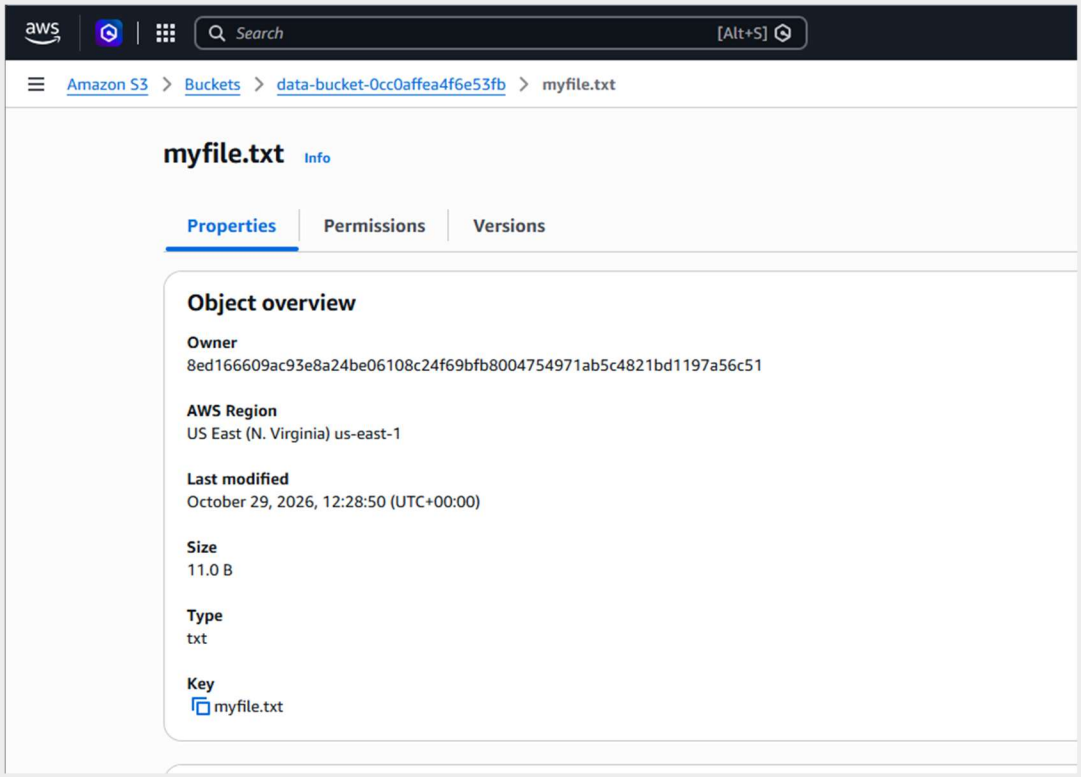


Figure 3 Paulo Access Authorised

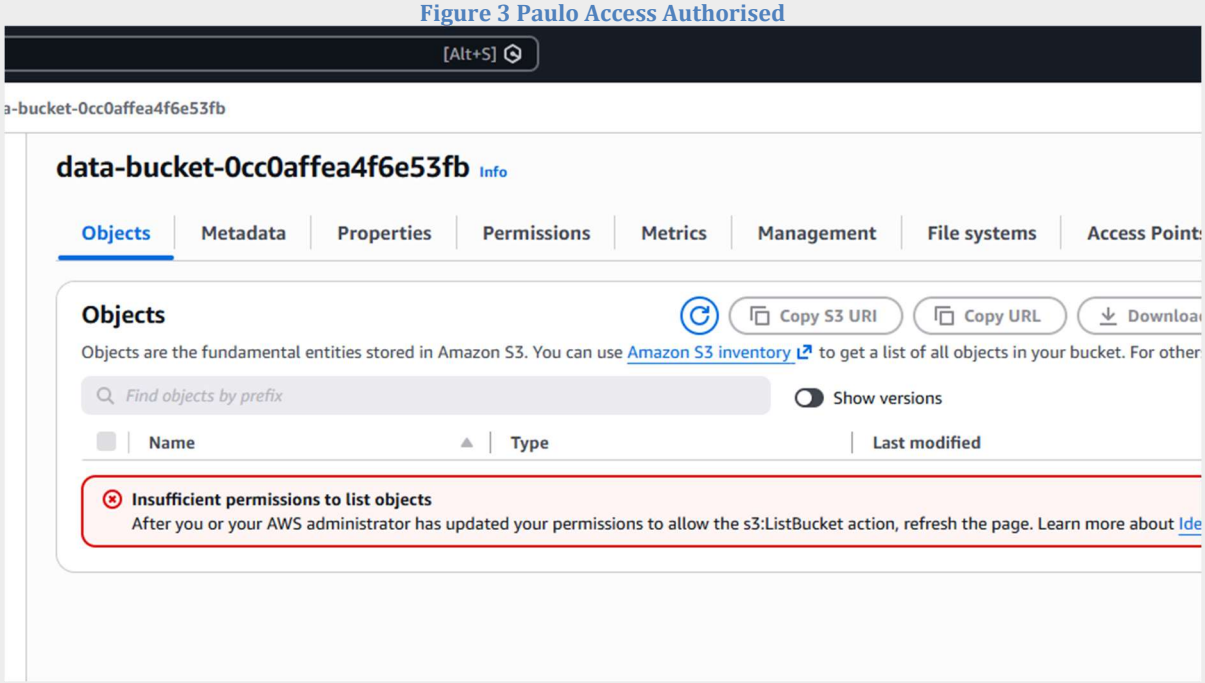
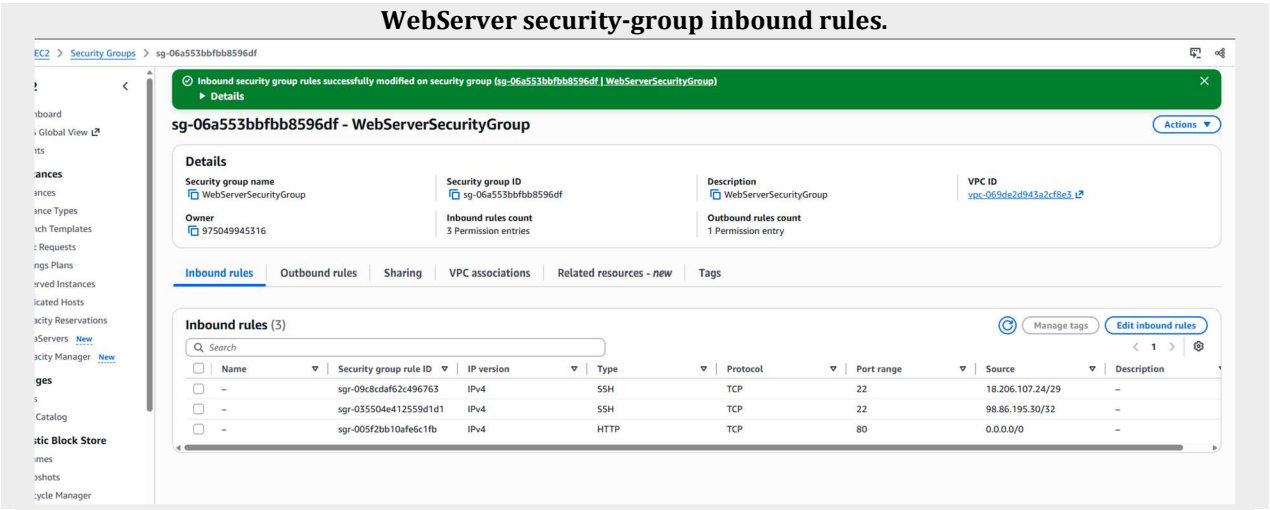


Figure 4 Mary access unauthorised

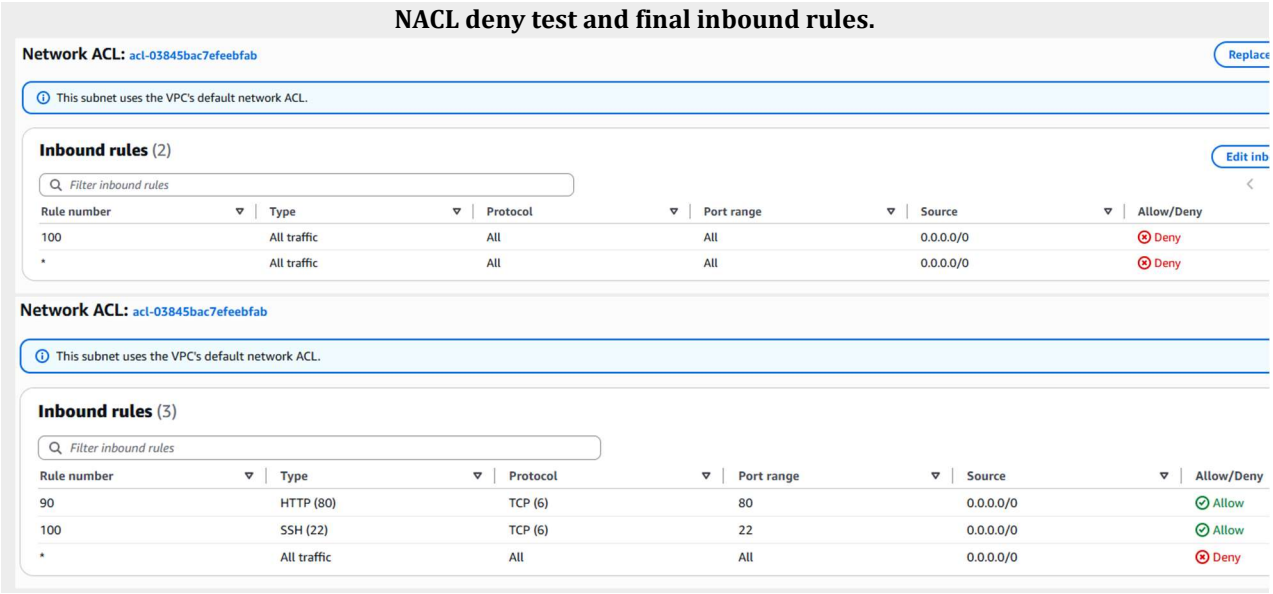
Phase 2 – Network Access

Network access was restricted at multiple layers. The security group permitted the required HTTP and authorised SSH traffic, the network ACL provided subnet-level filtering, and AWS Network Firewall applied stateful rules including an explicit drop for TCP port 8080.

WebServer security-group inbound rules.



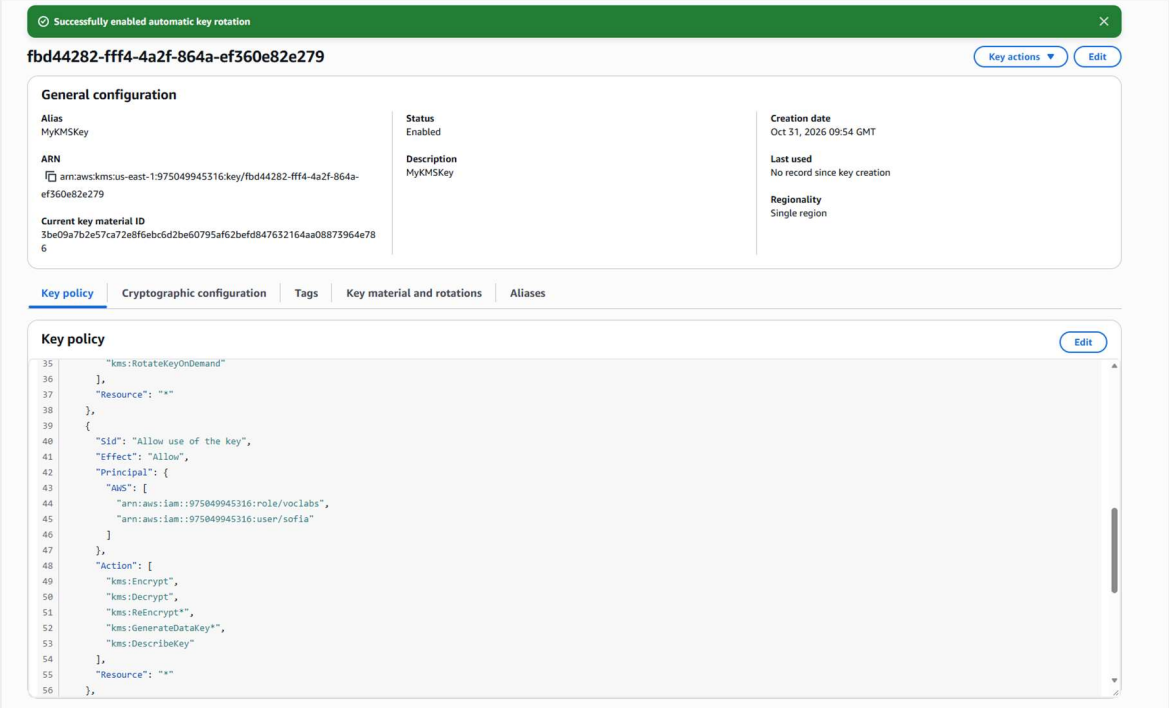
NACL deny test and final inbound rules.

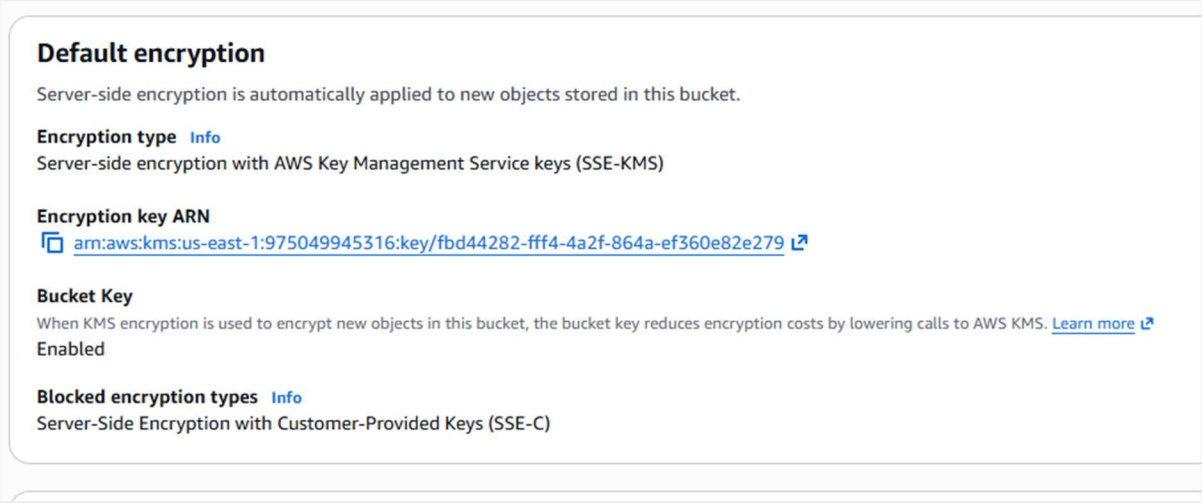


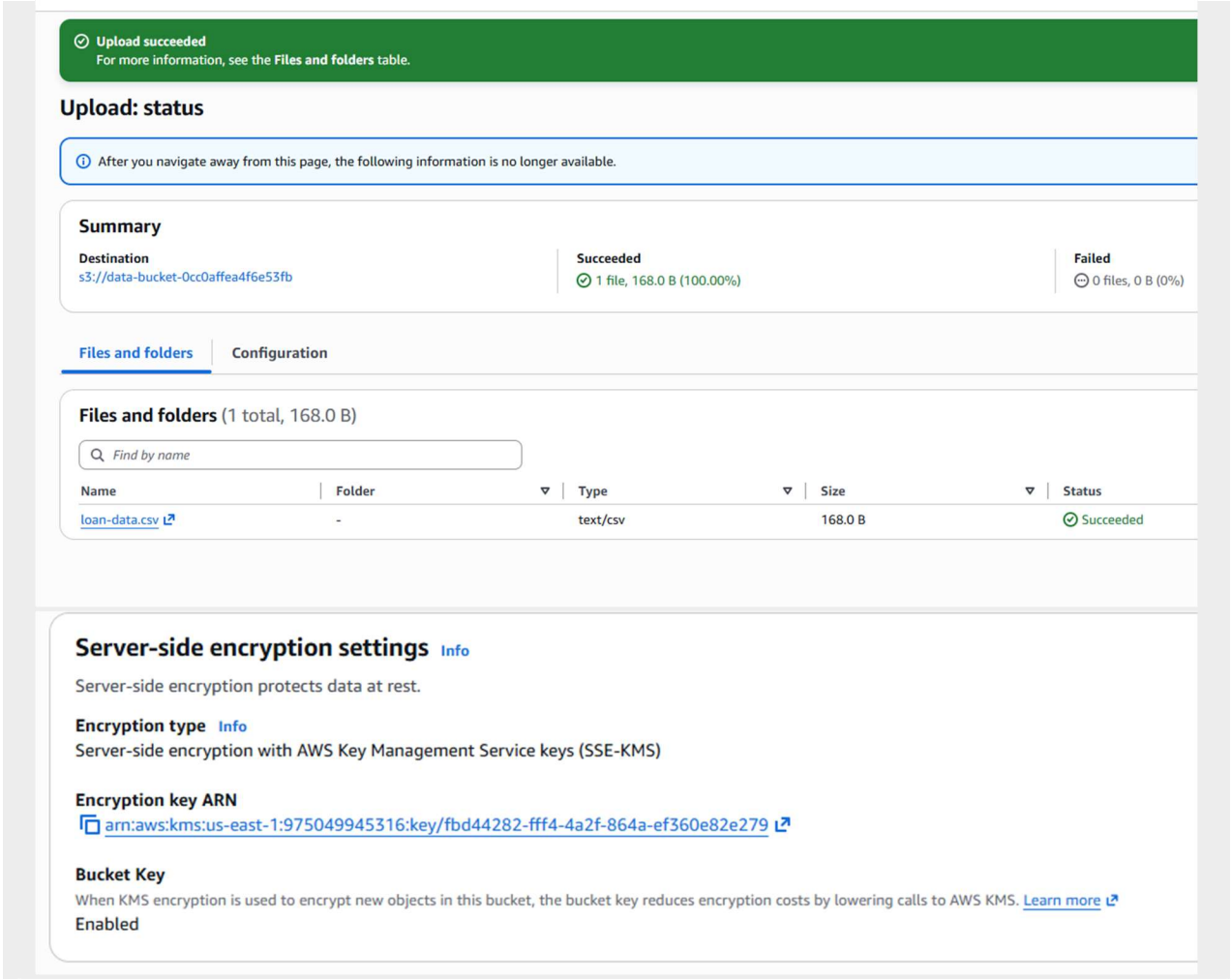
Phase 3 – IAM and KMS Access

Sofia was authorised to use MyKMSKey, whereas Paulo was not granted the same KMS permissions. Sofia could access the SSE-KMS encrypted loan-data.csv object, while Paulo's attempt failed. The result demonstrates that S3 access alone is insufficient when the user lacks permission to use the encryption key.

KMS rotation/key policy, encrypted S3 bucket, Sofia upload/encryption and Paulo failed download.







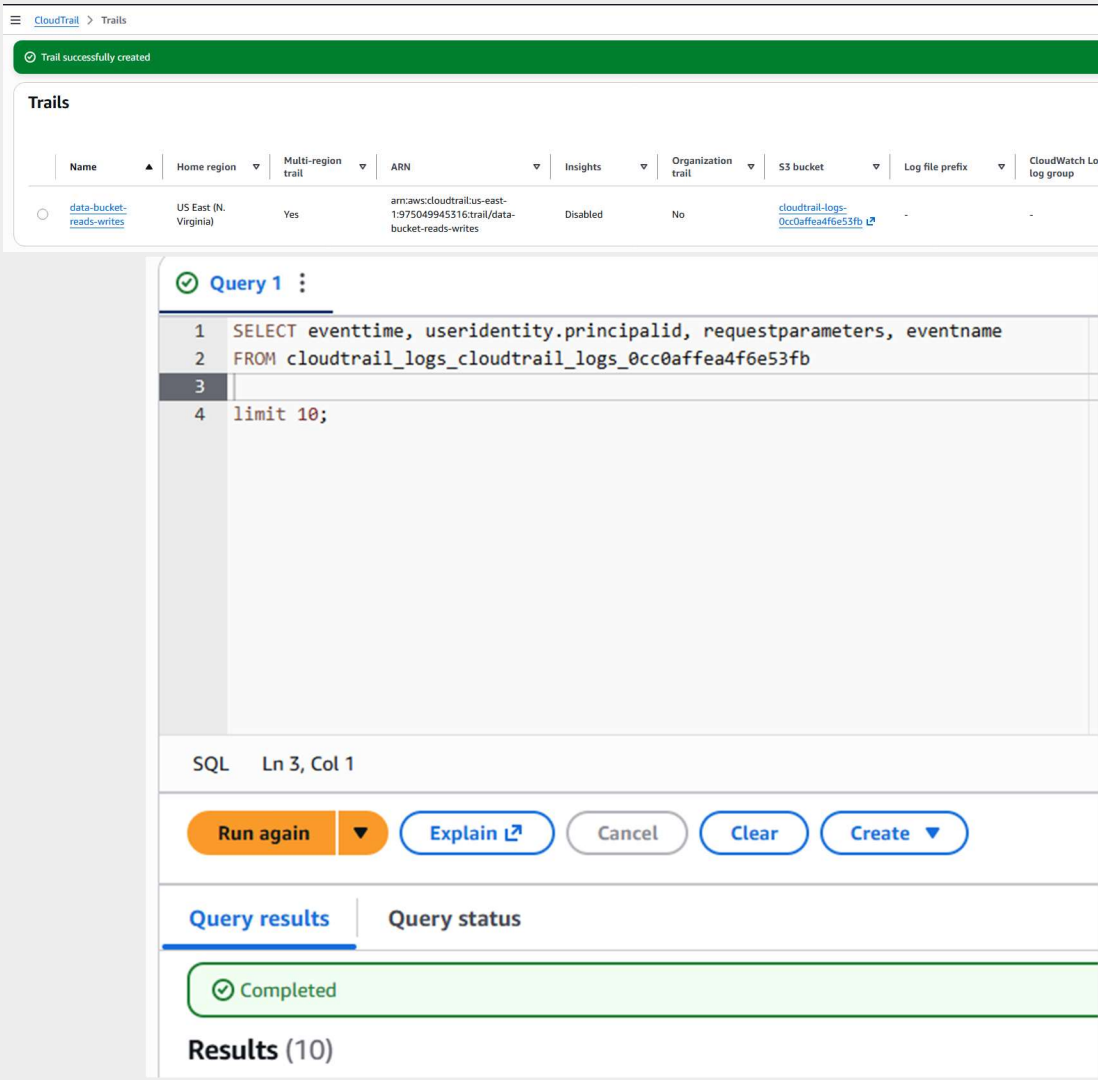
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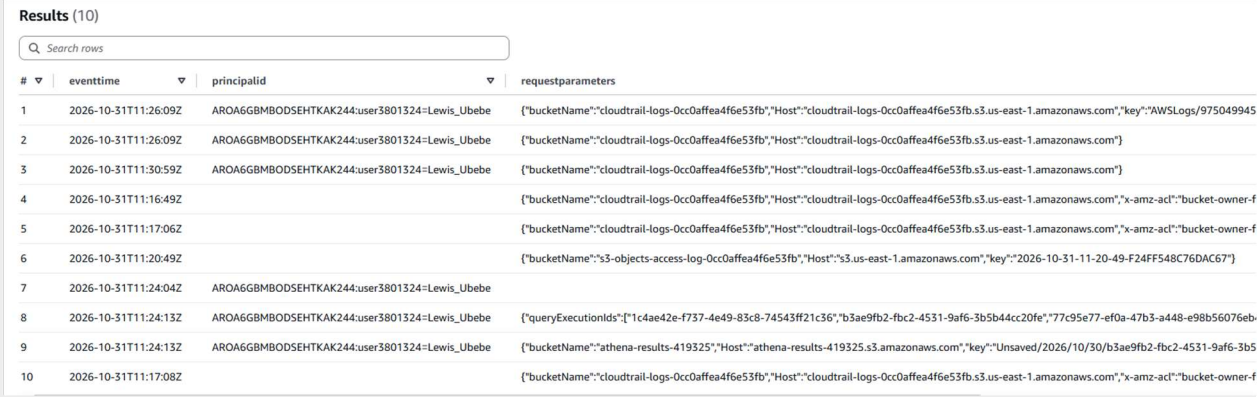

```
▼<Error>
   <Code>AccessDenied</Code>
   <Message>User: arn:aws:iam::975049945316:user/paulo is not authorized to perform: kms:Decrypt on resource: arn:aw
   <RequestId>D937MJSCHSSV0T11</RequestId>
   <HostId>xKjVoPx9GMJjslWI9n3dE4enA9SaVYE/3qlSpv4AjUlcYu9CKft8zneaQ/HXRRMqsqiuNcYYOrUBMBwIxfHf3hVunE6DgwHA</HostId>
 </Error>
```

R4 – Enforce Compliance

AWS Config was used to assess S3 logging compliance. The managed rule s3-bucket-logging-enabled identified the compliance bucket as noncompliant because server access logging was disabled. Remediation was then performed using AWS Systems Manager Automation, after which server access logging was enabled. CloudTrail/Athena evidence also provides an audit record of monitored S3 activity.

CloudTrail S3 read/write configuration, Athena query and verified results.

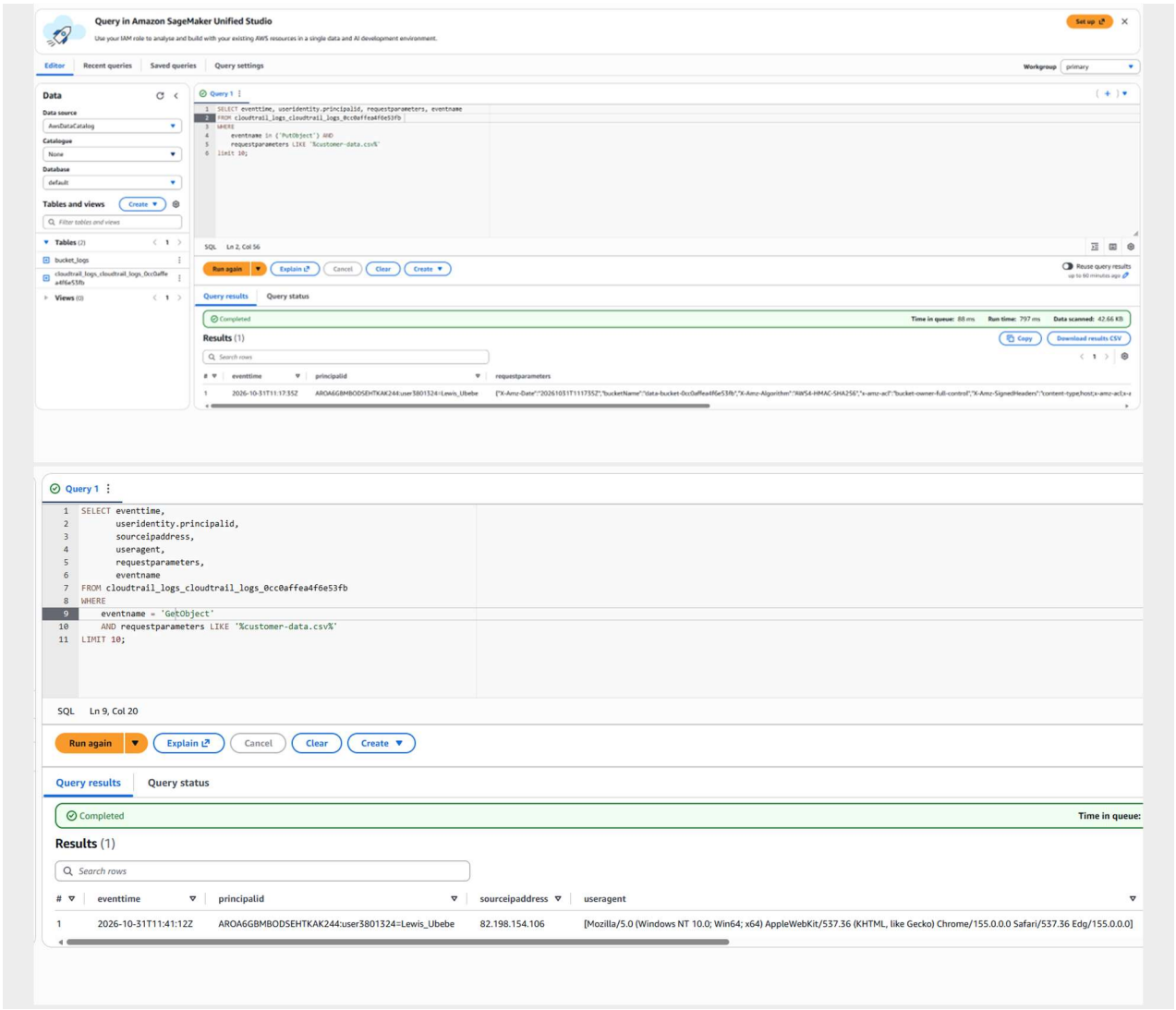




Results (10)

Search rows

#	eventtime	principalid	requestparameters
1	2026-10-31T11:26:09Z	AROA6GBMBODSEHTKAK244:user3801324=Lewis_Ubebe	{"bucketName":"cloudtrail-logs-0cc0affea4f6e53fb","Host":"cloudtrail-logs-0cc0affea4f6e53fb.s3.us-east-1.amazonaws.com","key":"AWSLogs/975049945
2	2026-10-31T11:26:09Z	AROA6GBMBODSEHTKAK244:user3801324=Lewis_Ubebe	{"bucketName":"cloudtrail-logs-0cc0affea4f6e53fb","Host":"cloudtrail-logs-0cc0affea4f6e53fb.s3.us-east-1.amazonaws.com"}
3	2026-10-31T11:30:59Z	AROA6GBMBODSEHTKAK244:user3801324=Lewis_Ubebe	{"bucketName":"cloudtrail-logs-0cc0affea4f6e53fb","Host":"cloudtrail-logs-0cc0affea4f6e53fb.s3.us-east-1.amazonaws.com"}
4	2026-10-31T11:16:49Z		{"bucketName":"cloudtrail-logs-0cc0affea4f6e53fb","Host":"cloudtrail-logs-0cc0affea4f6e53fb.s3.us-east-1.amazonaws.com","x-amz-acl":"bucket-owner-f
5	2026-10-31T11:17:06Z		{"bucketName":"cloudtrail-logs-0cc0affea4f6e53fb","Host":"cloudtrail-logs-0cc0affea4f6e53fb.s3.us-east-1.amazonaws.com","x-amz-acl":"bucket-owner-f
6	2026-10-31T11:20:49Z		{"bucketName":"s3-objects-access-log-0cc0affea4f6e53fb","Host":"s3.us-east-1.amazonaws.com","key":"2026-10-31-11-20-49-F24FF548C76DAC67"}
7	2026-10-31T11:24:04Z	AROA6GBMBODSEHTKAK244:user3801324=Lewis_Ubebe	
8	2026-10-31T11:24:13Z	AROA6GBMBODSEHTKAK244:user3801324=Lewis_Ubebe	{"queryExecutionIds":["1c4ae42e-f737-4e49-83c8-74543ff21c36","b3ae9fb2-fbc2-4531-9af6-3b5b44cc20fe","77c95e77-ef0a-47b3-a448-e98b56076eb
9	2026-10-31T11:24:13Z	AROA6GBMBODSEHTKAK244:user3801324=Lewis_Ubebe	{"bucketName":"athena-results-419325","Host":"athena-results-419325.s3.amazonaws.com","key":"Unsaved/2026/10/30/b3ae9fb2-fbc2-4531-9af6-3b5
10	2026-10-31T11:17:08Z		{"bucketName":"cloudtrail-logs-0cc0affea4f6e53fb","Host":"cloudtrail-logs-0cc0affea4f6e53fb.s3.us-east-1.amazonaws.com","x-amz-acl":"bucket-owner-f



AWS Config / S3 logging evidence showing the compliance control or noncompliant state.

Remediation action

Remediation action
AWS-ConfigureS3BucketLogging

Description
Enables Logging on S3 Bucket

Parameters

Key	Value	Description
AutomationAssumeRole	arn:aws:iam::975049945316:role/SSMAutomationRole	(Optional) The ARN of the role that allows Automation to perform the actions on your b
BucketName	RESOURCE_ID	(Required) The name of the Amazon S3 Bucket for which you want to configure logging
GrantedPermission	FULL_CONTROL	(Optional) Logging permissions assigned to the Grantee for the bucket.
GranteeEmailAddress	-	(Optional) Email address of the grantee.
GranteeId	8ed166609ac93e8a24be06108c24f69bfb8004754971ab5c4821bd1197a56c51	(Optional) The canonical user ID of the grantee.
GranteeType	CanonicalUser	(Optional) Type of grantee
GranteeUri	-	(Optional) URI of the grantee group.
TargetBucket	s3-objects-access-log-0cc0affea4f6e53fb	(Required) Specifies the bucket where you want Amazon S3 to store server access logs.
TargetObjectKeyPartitionDateSource	-	(Optional) Specifies the partition date source for the partitioned prefix.
TargetObjectKeyPrefix	-	(Optional) Amazon S3 key format for log objects.
TargetPrefix	-	(Optional) Specifies a prefix for the keys under which the log files will be stored.

Resource after remediation / compliant result.

Resources in scope

Noncompliant

ID	Type	Status
athena-results-419325	S3 Bucket	-
aws-athena-query-results-975049945316-us-east-1	S3 Bucket	-
aws-config-0cc0affea4f6e53fb	S3 Bucket	-
cloudtrail-logs-0cc0affea4f6e53fb	S3 Bucket	-
compliance-bucket-0cc0affea4f6e53fb	S3 Bucket	Action executed successfully
s3-inventory-0cc0affea4f6e53fb	S3 Bucket	-
s3-objects-access-log-0cc0affea4f6e53fb	S3 Bucket	-

Compliance bucket with server access logging enabled.

Server access logging

Log requests for access to your bucket. Use CloudWatch to check the health of your server access logging. Learn more

Server access logging
Enabled

Destination bucket
s3://s3-objects-access-log-0cc0affea4f6e53fb

Log object key format
/[YYYY]-[MM]-[DD]-[hh]-[mm]-[ss]-[UniqueString]

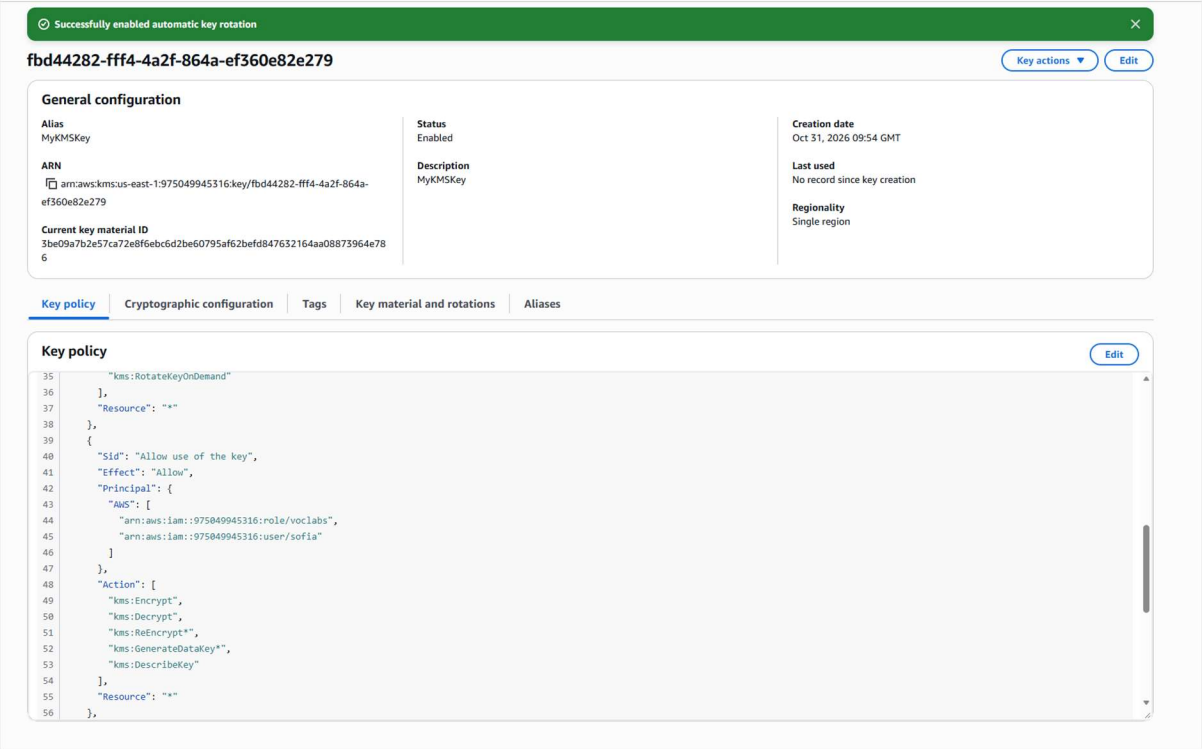
R5 – Encrypt Data

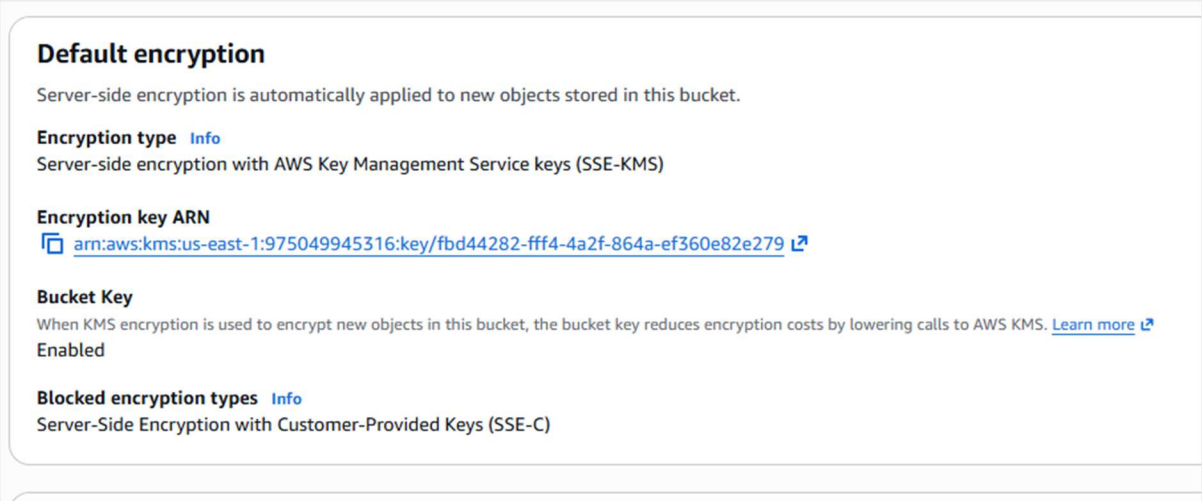
A customer-managed AWS KMS key, MyKMSKey, was used throughout Phase 3. Automatic key rotation was enabled and the key policy authorised Sofia to use the key.

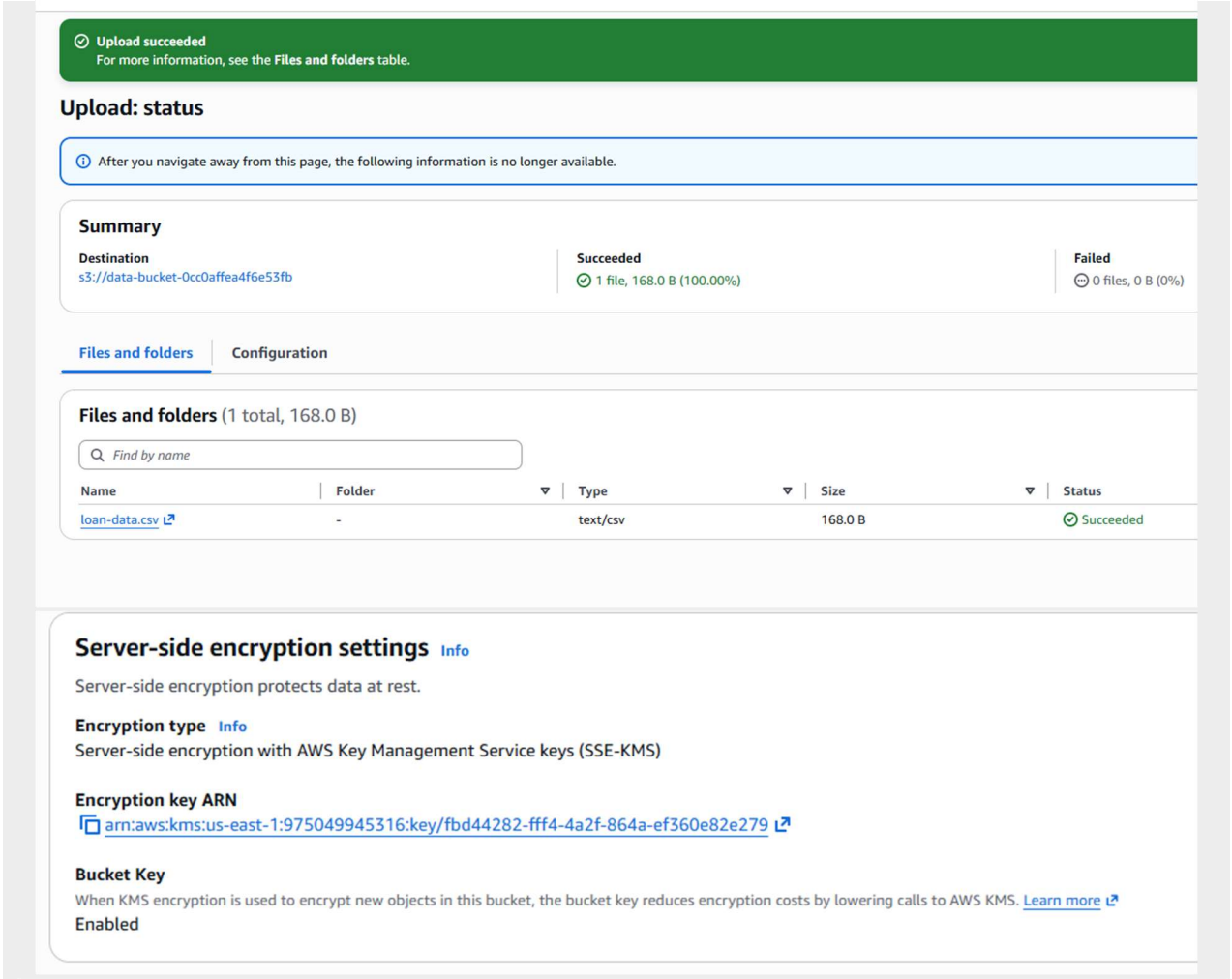
S3 – SSE-KMS

The data bucket and loan-data.csv object were protected with SSE-KMS.

KMS rotation/key policy, encrypted S3 bucket, Sofia upload and loan-data.csv SSE-KMS settings.







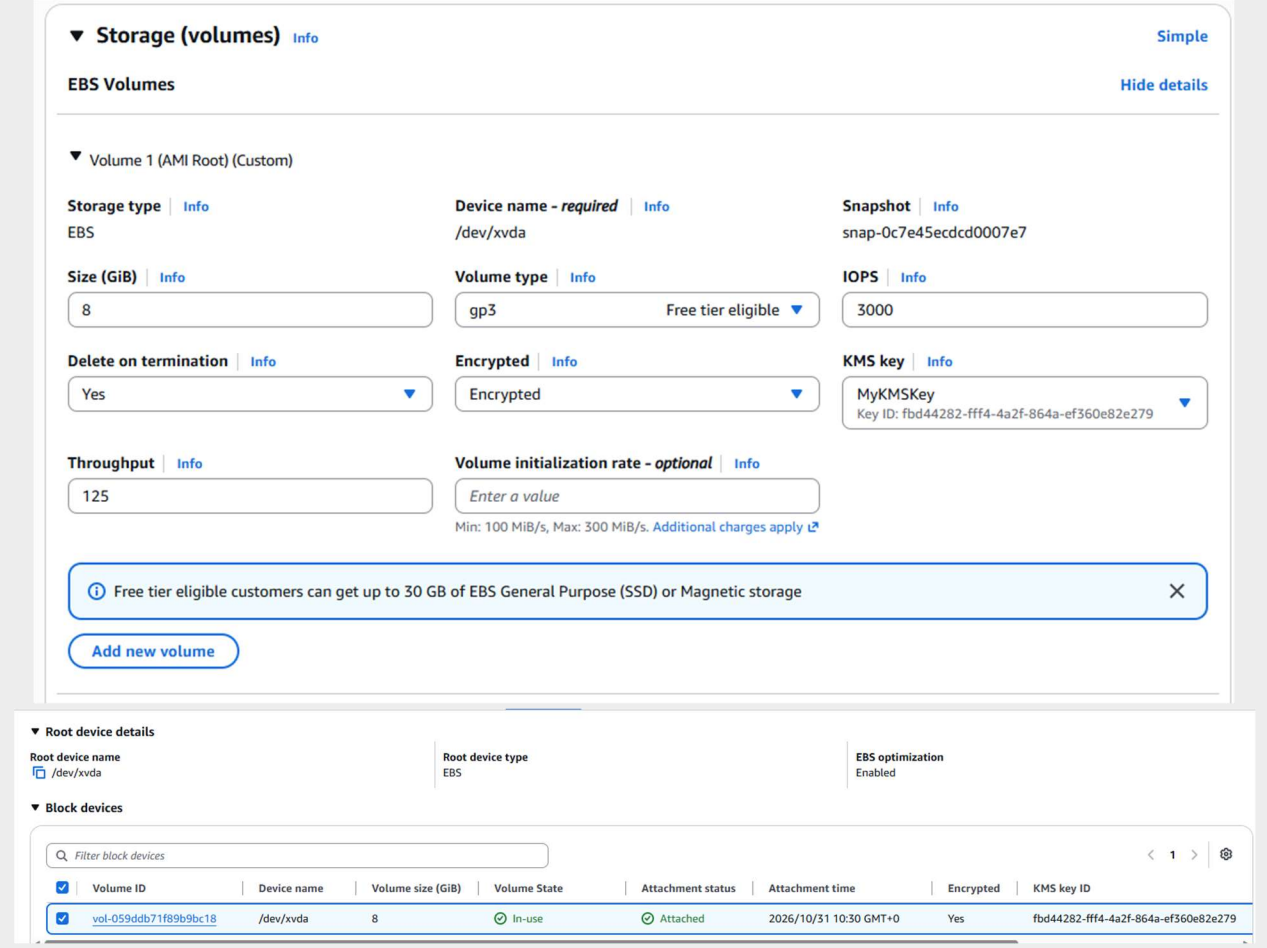
This XML file does not appear to have any style information associated with it. The document tree is shown below.


```
▼<Error>
   <Code>AccessDenied</Code>
   <Message>User: arn:aws:iam::975049945316:user/paulo is not authorized to perform: kms:Decrypt on resource: arn:a
   <RequestId>D937MJSCHSSV0T11</RequestId>
   <HostId>xKjVoPx9GMJjslWI9n3dE4enA9SaVYE/3qlSpv4AjUlcYu9CKft8zneaQ/HXRRMqsqiuNcYYOrUBMBwIxfHf3hVunE6DgwHA</HostId
 </Error>
```

EBS Root Volume Encryption

EncryptedInstance was created with its root EBS volume encrypted using MyKMSKey.

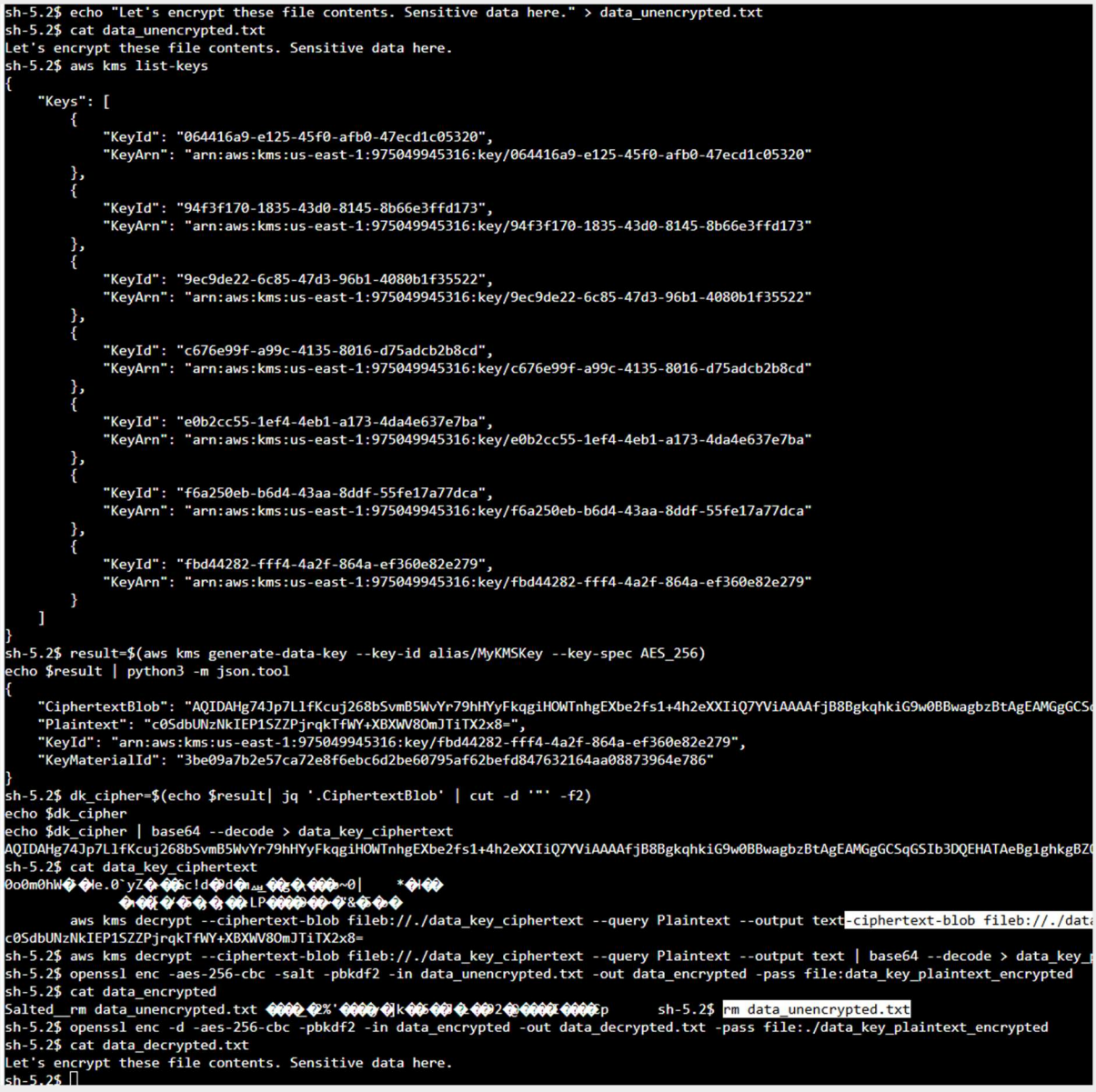
Encrypted EC2 configuration and encrypted root-device volume.



Envelope Encryption

Envelope encryption was demonstrated on WebServer2 by generating a KMS data key and using that data key to encrypt local data with AES-256. The encrypted file was unreadable until decrypted with the protected data key.

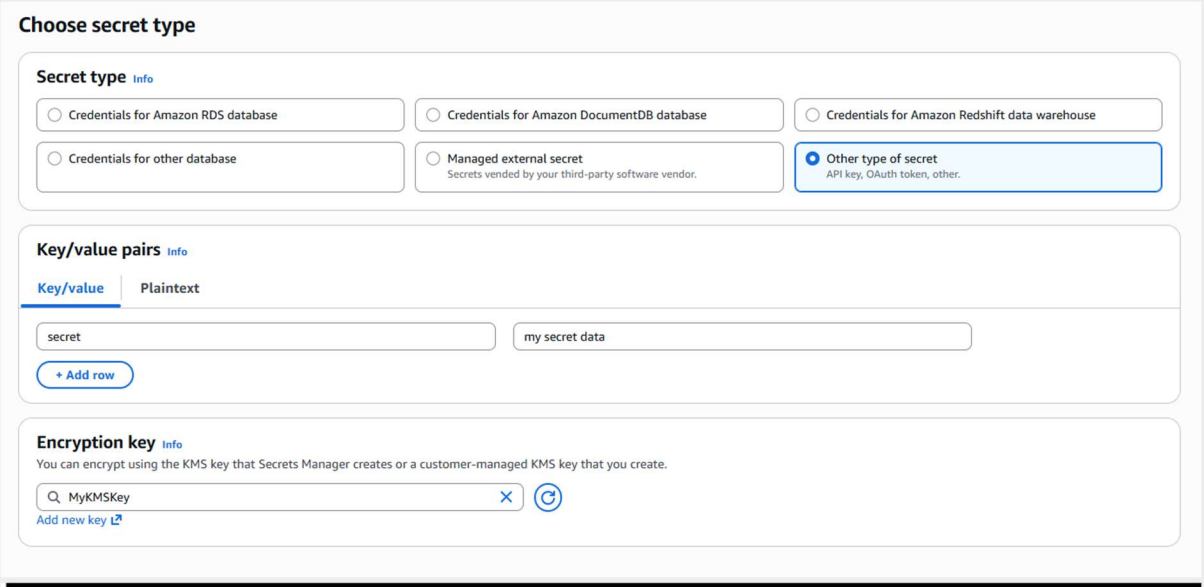
CLI evidence demonstrating envelope encryption and encrypted data at rest.



Secrets Manager

A secret was stored in AWS Secrets Manager using MyKMSKey and successfully retrieved from WebServer2 using the AWS CLI.

New secret configuration and successful CLI retrieval from WebServer2.

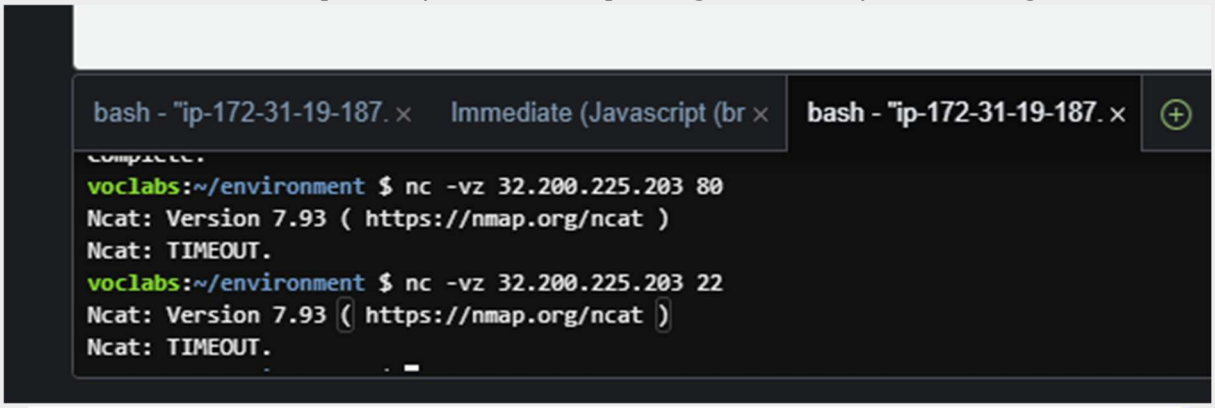


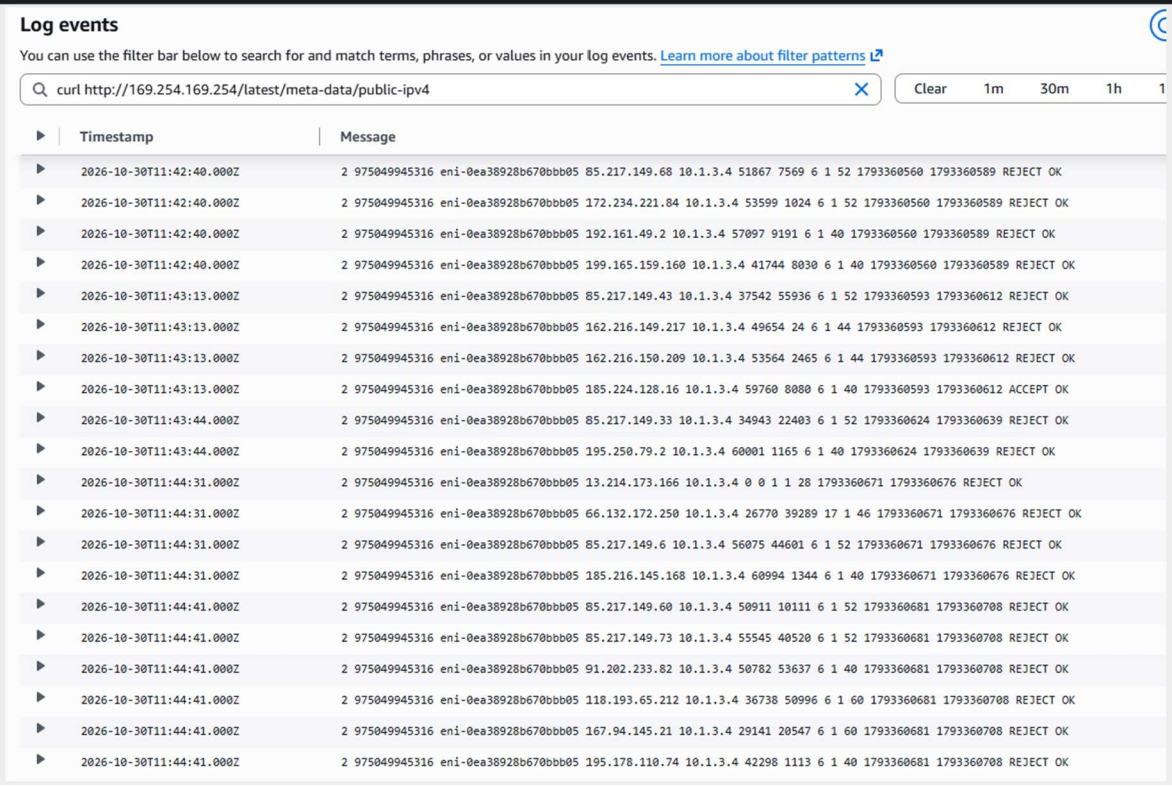

```
sh-5.2$ aws secretsmanager list-secrets
{
    "SecretList": [
        {
            "ARN": "arn:aws:secretsmanager:us-east-1:975049945316:secret:mysecret-gDwUQy",
            "Name": "mysecret",
            "KmsKeyId": "arn:aws:kms:us-east-1:975049945316:key/fbd44282-fff4-4a2f-864a-ef360e82e279",
            "LastChangedDate": "2026-10-31T10:47:09.844000+00:00",
            "Tags": [],
            "SecretVersionsToStages": {
                "0641ab51-1bb0-48d2-9eff-b10ff4b8f231": [
                    "AWSCURRENT"
                ]
            },
            "CreatedDate": "2026-10-31T10:47:09.765000+00:00"
        }
    ]
}
sh-5.2$ aws secretsmanager get-secret-value --secret-id mysecret
{
    "ARN": "arn:aws:secretsmanager:us-east-1:975049945316:secret:mysecret-gDwUQy",
    "Name": "mysecret",
    "VersionId": "0641ab51-1bb0-48d2-9eff-b10ff4b8f231",
    "SecretString": "{\"secret\":\"my secret data\"}",
    "VersionStages": [
        "AWSCURRENT"
    ],
    "CreatedDate": "2026-10-31T10:47:09.839000+00:00"
}
sh-5.2$
```

R6 – WITHSTAND PENETRATION TESTING

Connectivity tests were performed using Netcat, browser requests, EC2 Instance Connect and ICMP. Initially, ports 22 and 80 failed because the network was incorrectly configured. After the route and security-group changes, the authorised ports became reachable. The NACL deny test then proved that subnet-level controls could block traffic even when the security group allowed it.

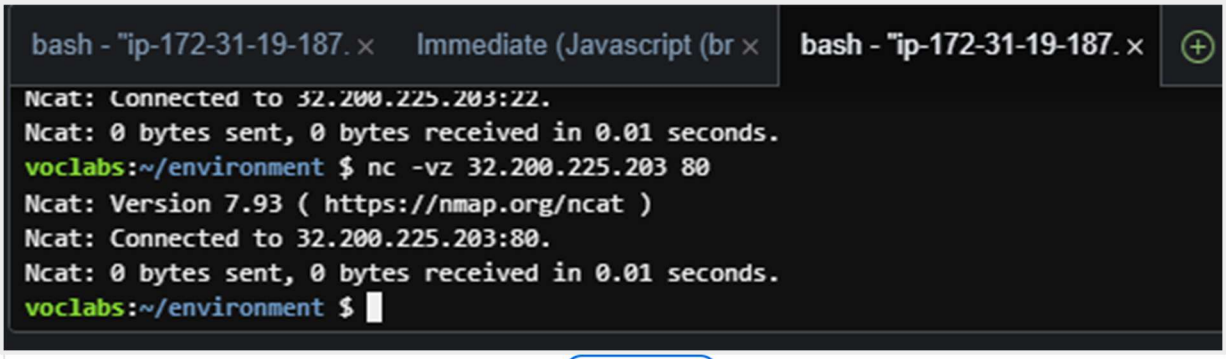
Initial Netcat tests on ports 80/22 and corresponding CloudWatch/VPC Flow Log evidence.

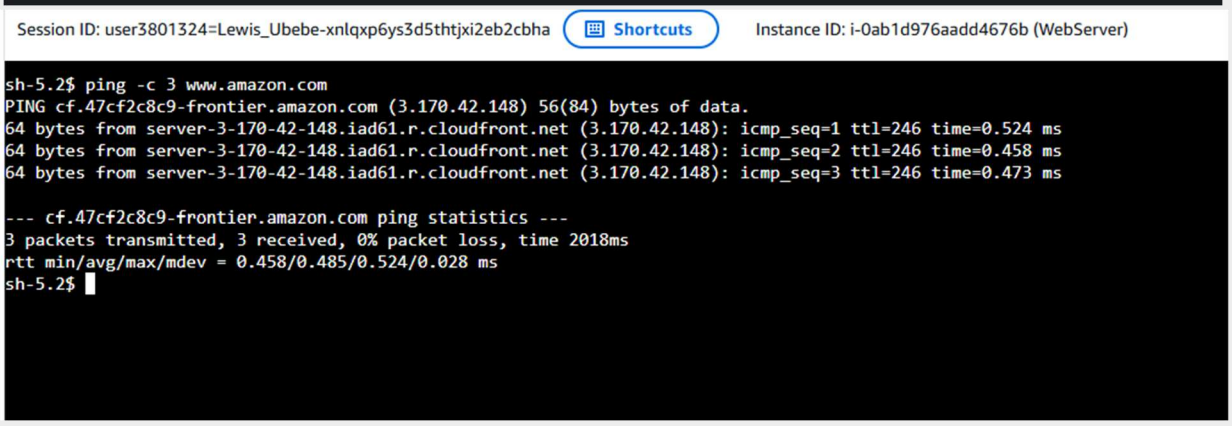


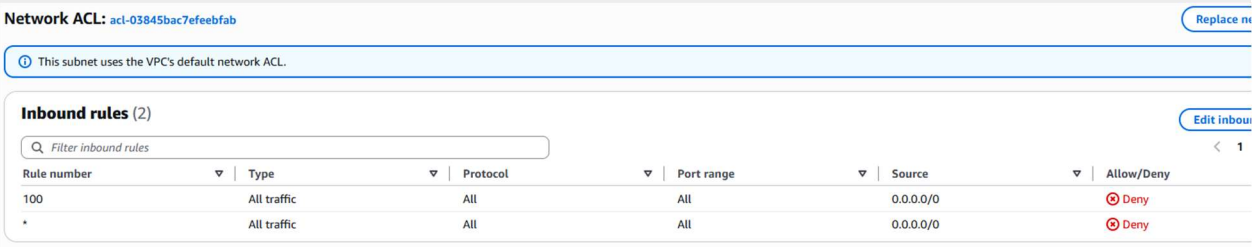


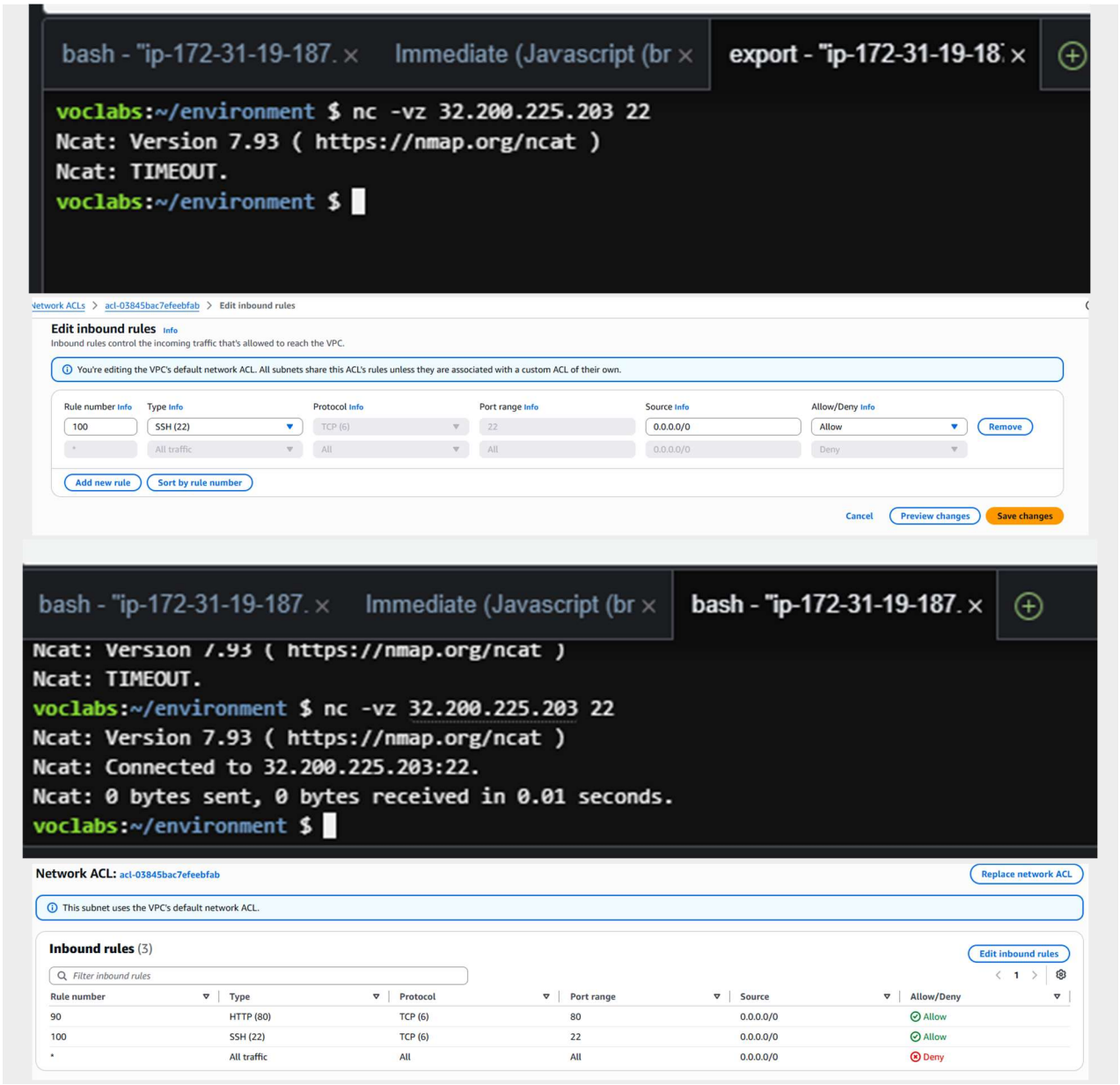
Successful authorised connections, NACL deny test, timeout and final NACL rules.





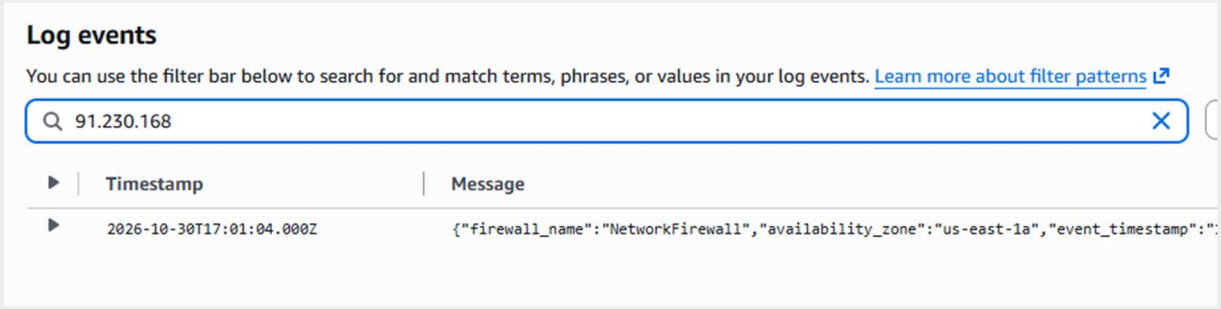


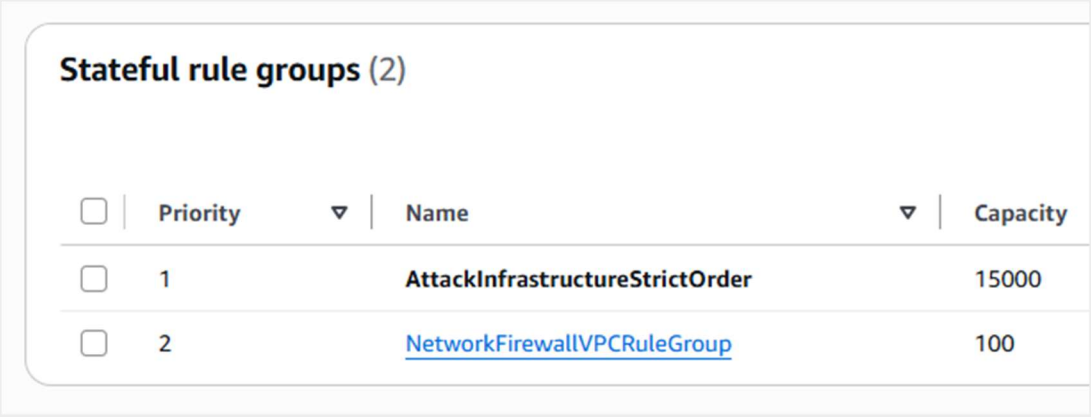


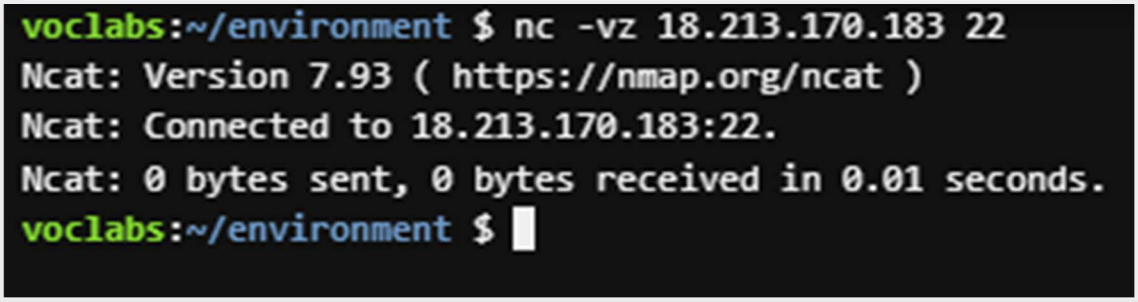


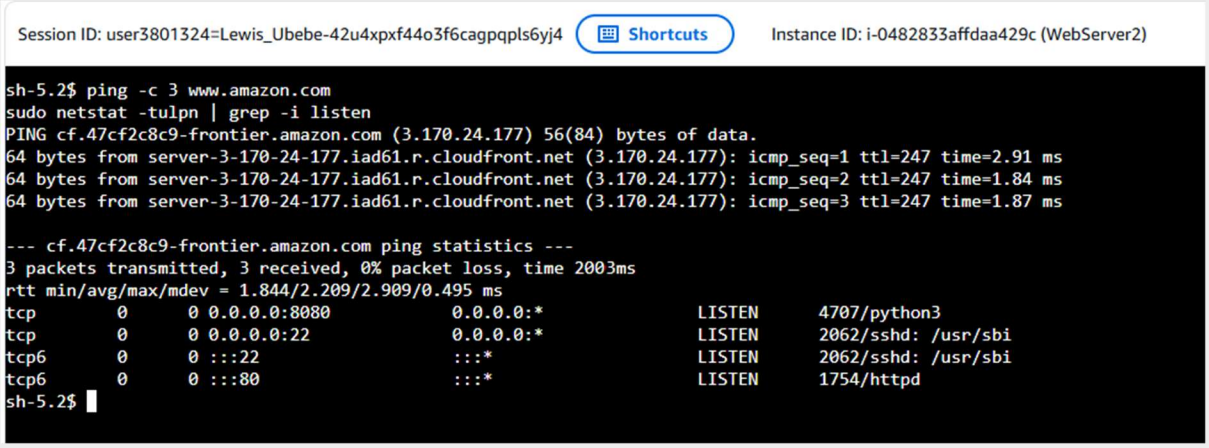
AWS Network Firewall was then tested with the final stateful policy. TCP ports 80 and 22 and ICMP were successfully used, while TCP port 8080 was blocked. Firewall logs captured the test traffic, demonstrating that only explicitly permitted services remained reachable.

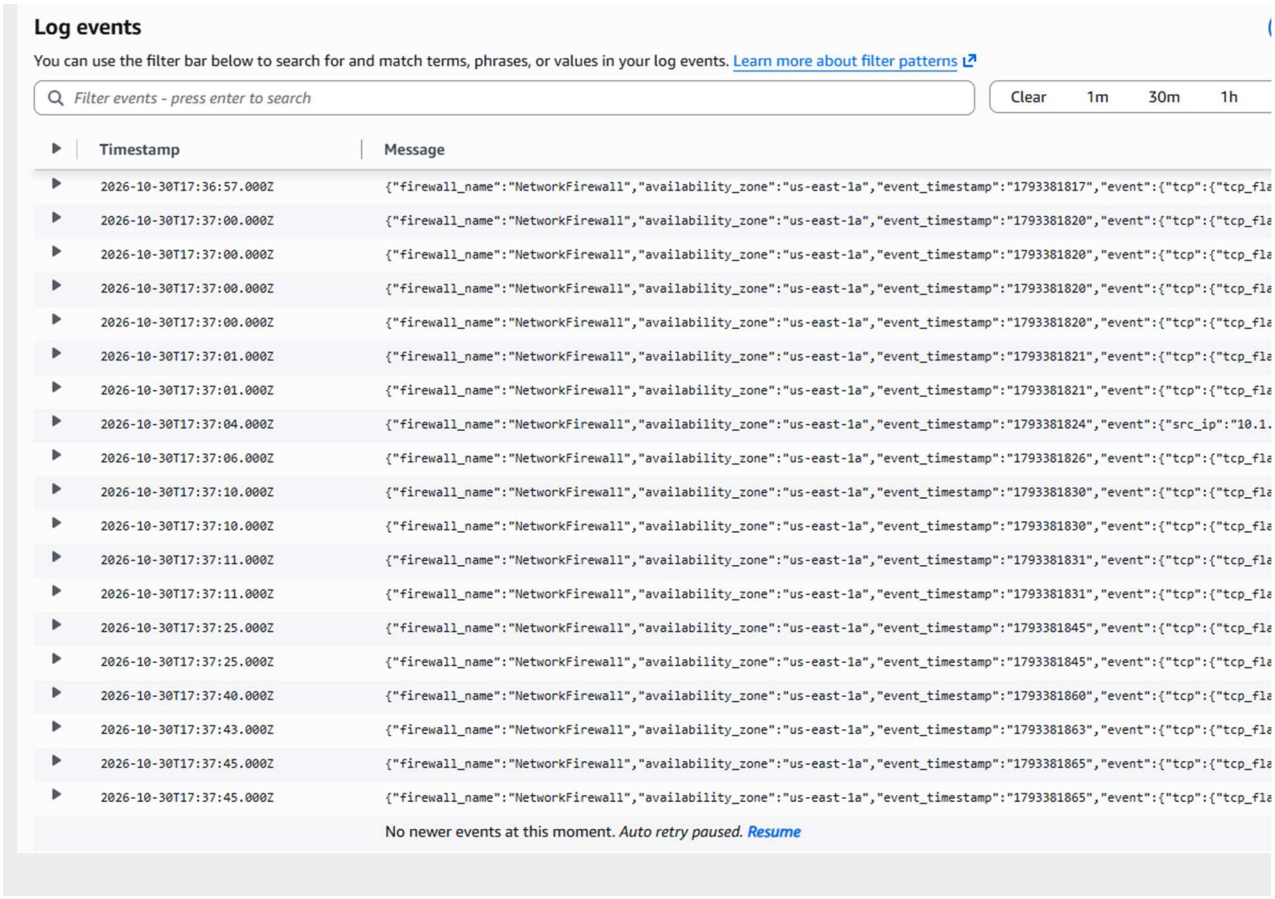
Firewall timeout/log event, final rule group, SSH success, ping success and CloudWatch firewall logs.









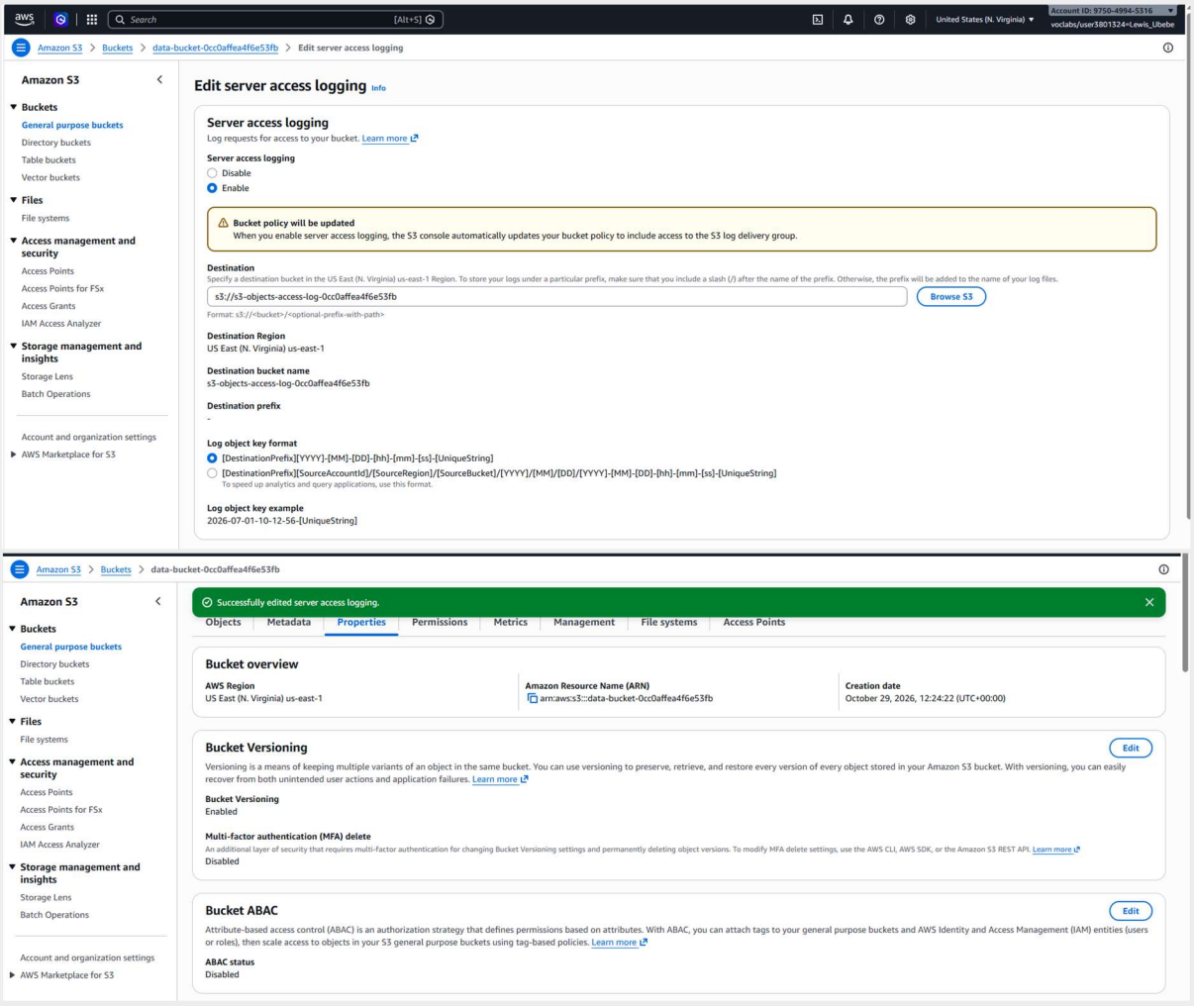

Log events
You can use the filter bar below to search for and match terms, phrases, or values in your log events. Learn more about filter patterns
Filter events - press enter to search
Clear
1m
30m
1h
Timestamp
Message
2026-10-30T17:36:57.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381817","event":{"tcp":{"tcp_fla
2026-10-30T17:37:00.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381820","event":{"tcp":{"tcp_fla
2026-10-30T17:37:00.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381820","event":{"tcp":{"tcp_fla
2026-10-30T17:37:00.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381820","event":{"tcp":{"tcp_fla
2026-10-30T17:37:00.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381820","event":{"tcp":{"tcp_fla
2026-10-30T17:37:01.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381821","event":{"tcp":{"tcp_fla
2026-10-30T17:37:01.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381821","event":{"tcp":{"tcp_fla
2026-10-30T17:37:04.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381824","event":{"src_ip":"10.1.
2026-10-30T17:37:06.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381826","event":{"tcp":{"tcp_fla
2026-10-30T17:37:10.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381830","event":{"tcp":{"tcp_fla
2026-10-30T17:37:10.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381830","event":{"tcp":{"tcp_fla
2026-10-30T17:37:11.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381831","event":{"tcp":{"tcp_fla
2026-10-30T17:37:11.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381831","event":{"tcp":{"tcp_fla
2026-10-30T17:37:25.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381845","event":{"tcp":{"tcp_fla
2026-10-30T17:37:25.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381845","event":{"tcp":{"tcp_fla
2026-10-30T17:37:40.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381860","event":{"tcp":{"tcp_fla
2026-10-30T17:37:43.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381863","event":{"tcp":{"tcp_fla
2026-10-30T17:37:45.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381865","event":{"tcp":{"tcp_fla
2026-10-30T17:37:45.000Z {"firewall_name":"NetworkFirewall","availability_zone":"us-east-1a","event_timestamp":"1793381865","event":{"tcp":{"tcp_fla
No newer events at this moment. Auto retry paused. Resume

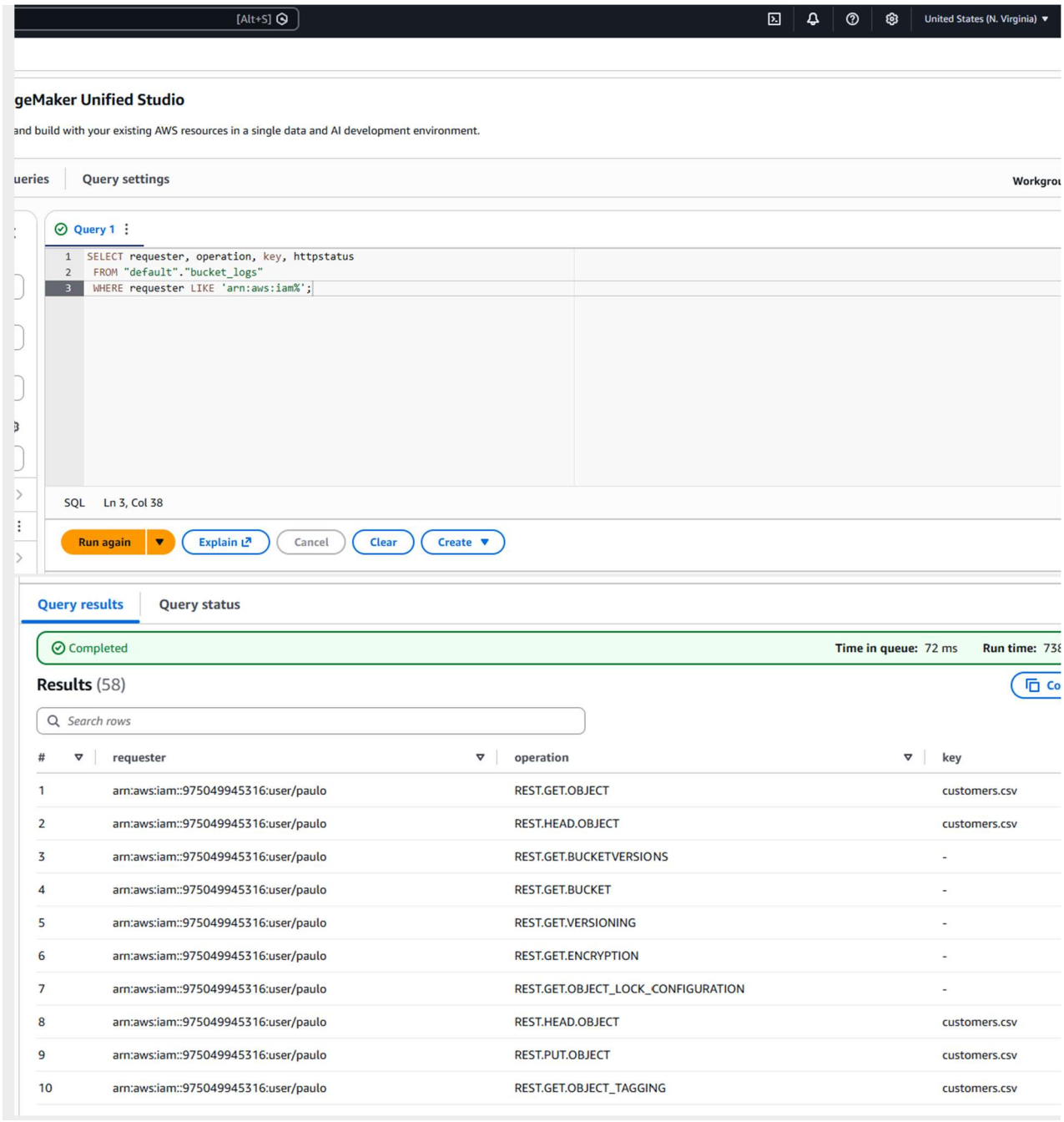
R7 – Monitor and Log User Activity

S3 Object Logging

Server access logging was enabled for the S3 data bucket and the resulting logs were queried with Amazon Athena. This created an audit trail for object access and allowed successful and denied activity to be analysed.

S3 server access logging, Athena access-log query and results.



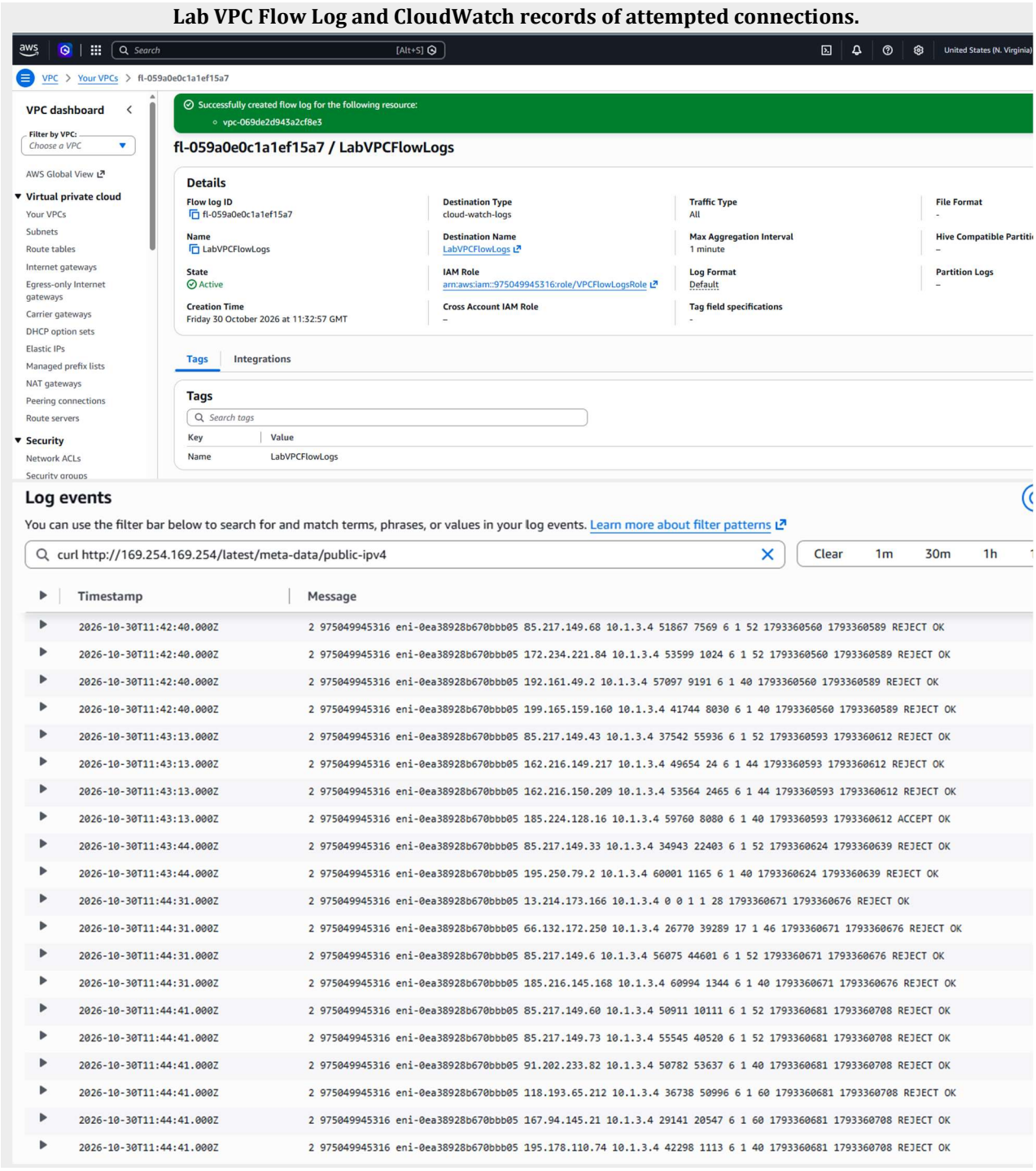


#	requester	operation	key
1	arn:aws:iam::975049945316:user/paulo	REST.GET.OBJECT	customers.csv
2	arn:aws:iam::975049945316:user/paulo	REST.HEAD.OBJECT	customers.csv
3	arn:aws:iam::975049945316:user/paulo	REST.GET.BUCKETVERSIONS	-
4	arn:aws:iam::975049945316:user/paulo	REST.GET.BUCKET	-
5	arn:aws:iam::975049945316:user/paulo	REST.GET.VERSIONING	-
6	arn:aws:iam::975049945316:user/paulo	REST.GET.ENCRYPTION	-
7	arn:aws:iam::975049945316:user/paulo	REST.GET.OBJECT_LOCK_CONFIGURATION	-
8	arn:aws:iam::975049945316:user/paulo	REST.HEAD.OBJECT	customers.csv
9	arn:aws:iam::975049945316:user/paulo	REST.PUT.OBJECT	customers.csv
10	arn:aws:iam::975049945316:user/paulo	REST.GET.OBJECT_TAGGING	customers.csv

VPC Flow Logs

LabVPC flow logging captured network traffic and delivered the records to CloudWatch Logs, including the Netcat connection attempts used during network testing.

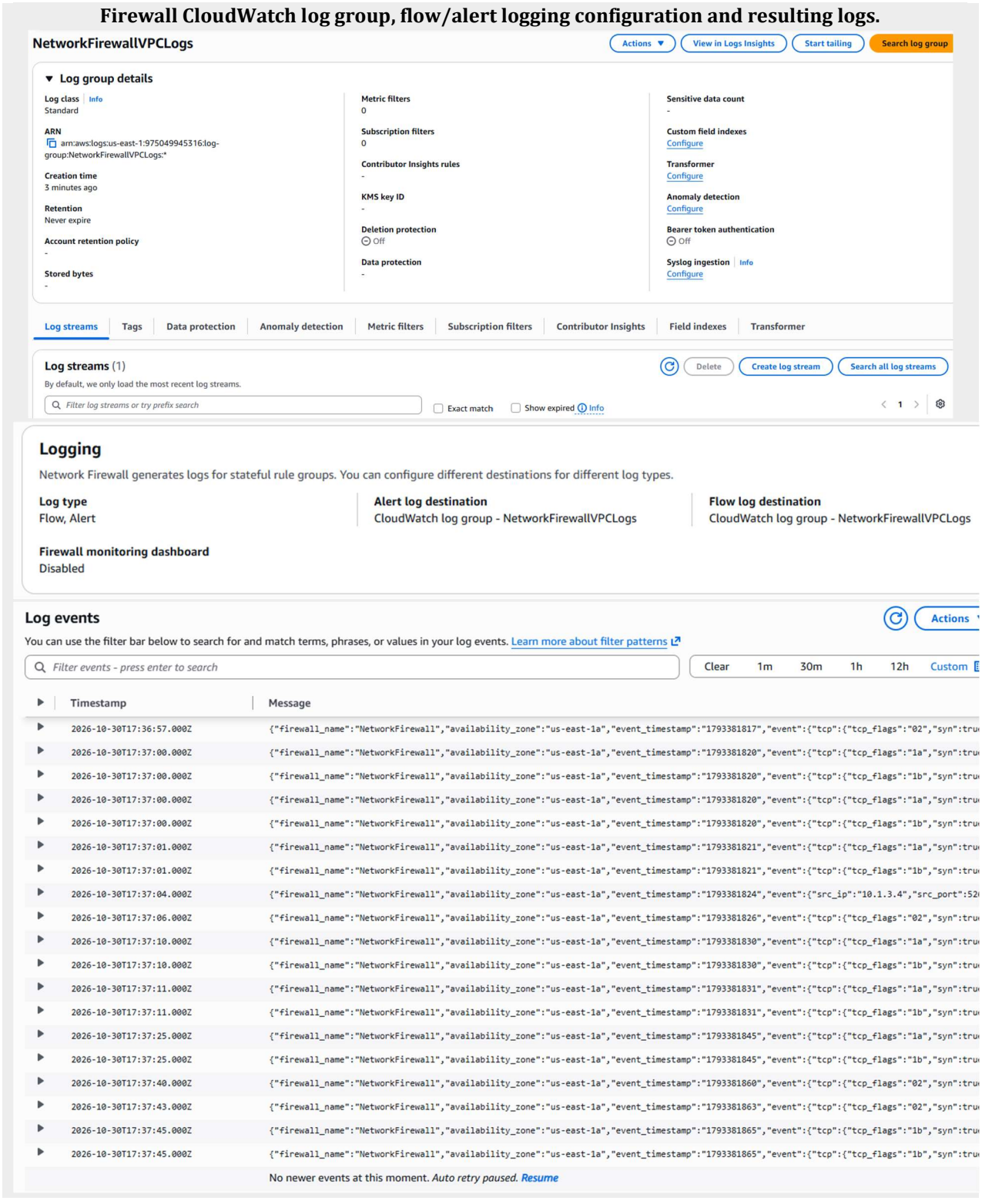
Lab VPC Flow Log and CloudWatch records of attempted connections.



Network Firewall Logging

AWS Network Firewall flow and alert logging was sent to the NetworkFirewallVPCLogs CloudWatch log group.

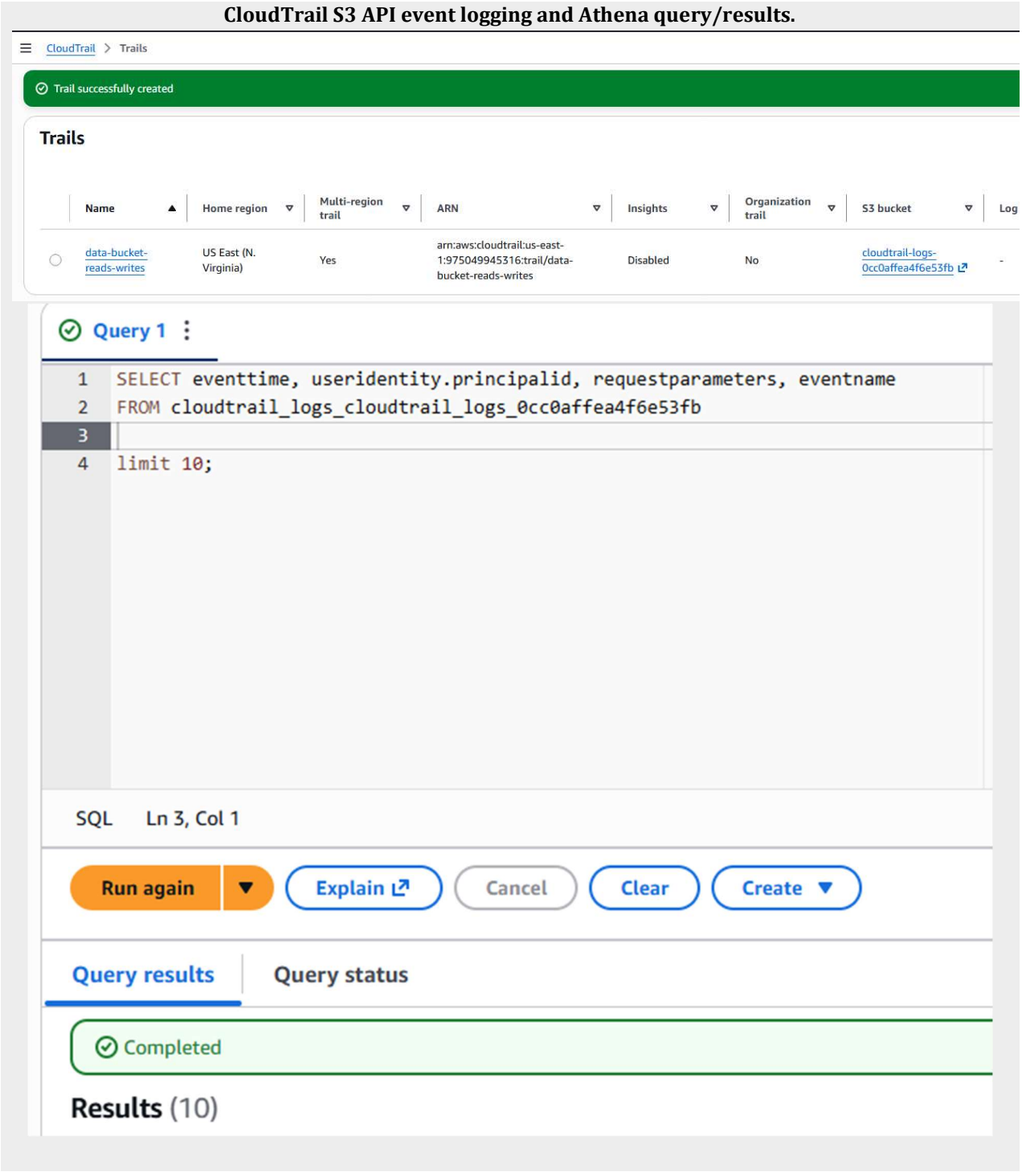
Firewall CloudWatch log group, flow/alert logging configuration and resulting logs.



CloudTrail and CloudWatch Security Monitoring

CloudTrail recorded S3 API actions and Athena was used to query the resulting event data. Secure log monitoring was also configured for the EC2 environment, and an unauthorised username attempt produced a permission-denied event.

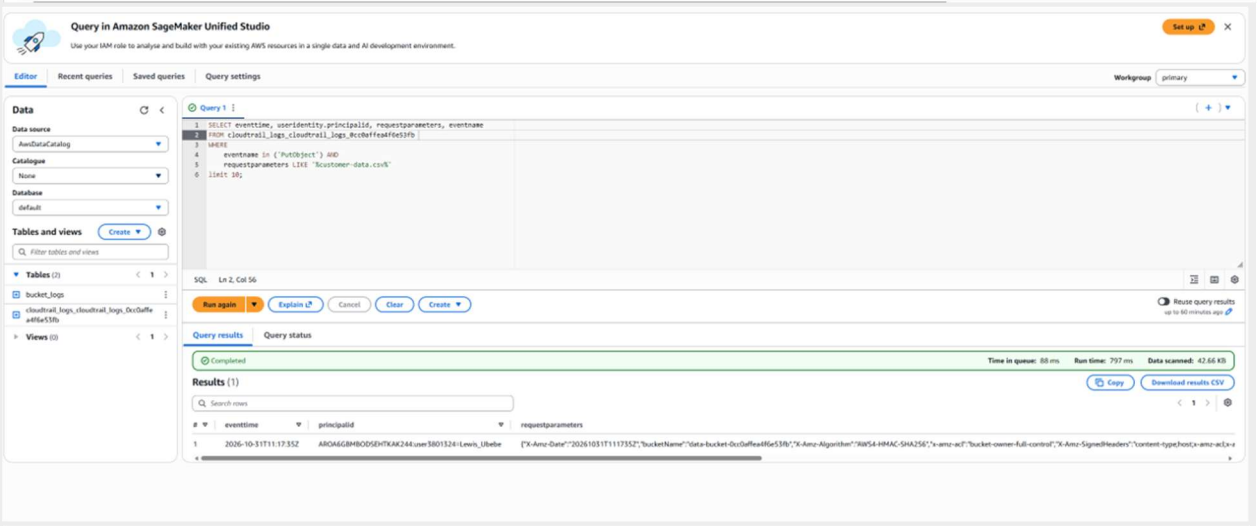
CloudTrail S3 API event logging and Athena query/results.

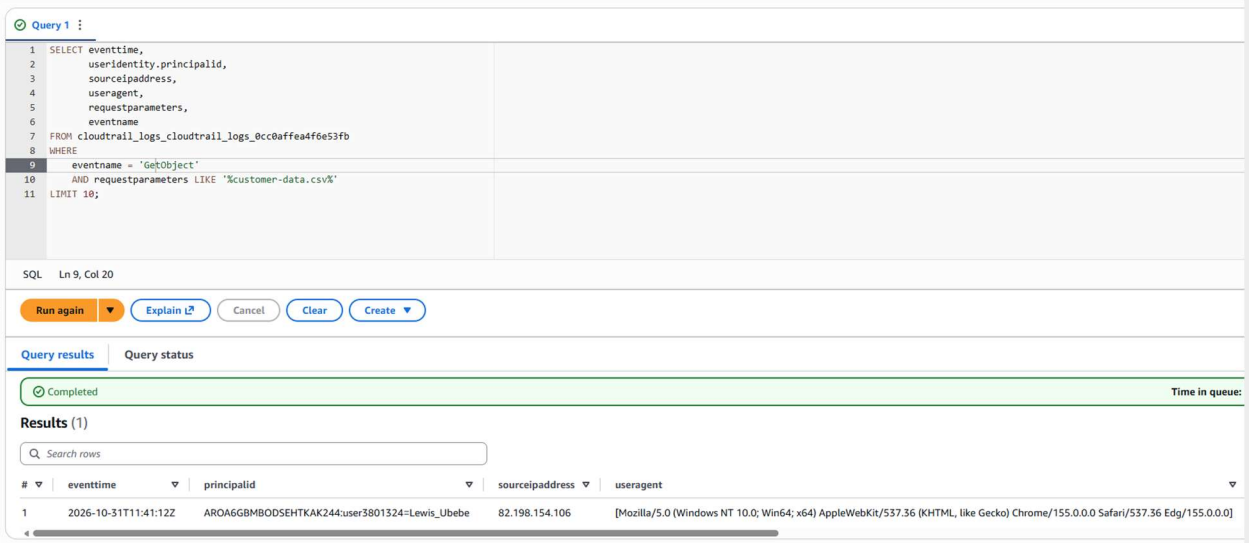


Results (10)

Search rows

#	eventtime	principalid	requestparameters
1	2026-10-31T11:26:09Z	AROA6GBMBODSEHTKAK244:user3801324=Lewis_Ubebe	{"bucketName":"cloudtrail-logs-0cc0affea4f6e53fb","Host":"cloudtrail-logs-0cc0affea4f6e53fb.s3.us-east-1.amazonaws.com","key":"AWSLogs/975049945316/Clou
2	2026-10-31T11:26:09Z	AROA6GBMBODSEHTKAK244:user3801324=Lewis_Ubebe	{"bucketName":"cloudtrail-logs-0cc0affea4f6e53fb","Host":"cloudtrail-logs-0cc0affea4f6e53fb.s3.us-east-1.amazonaws.com"}
3	2026-10-31T11:30:59Z	AROA6GBMBODSEHTKAK244:user3801324=Lewis_Ubebe	{"bucketName":"cloudtrail-logs-0cc0affea4f6e53fb","Host":"cloudtrail-logs-0cc0affea4f6e53fb.s3.us-east-1.amazonaws.com"}
4	2026-10-31T11:16:49Z		{"bucketName":"cloudtrail-logs-0cc0affea4f6e53fb","Host":"cloudtrail-logs-0cc0affea4f6e53fb.s3.us-east-1.amazonaws.com","x-amz-acl":"bucket-owner-full-contro
5	2026-10-31T11:17:06Z		{"bucketName":"cloudtrail-logs-0cc0affea4f6e53fb","Host":"cloudtrail-logs-0cc0affea4f6e53fb.s3.us-east-1.amazonaws.com","x-amz-acl":"bucket-owner-full-contro
6	2026-10-31T11:20:49Z		{"bucketName":"s3-objects-access-log-0cc0affea4f6e53fb","Host":"s3.us-east-1.amazonaws.com","key":"2026-10-31-11-20-49-F24FF548C76DAC67"}
7	2026-10-31T11:24:04Z	AROA6GBMBODSEHTKAK244:user3801324=Lewis_Ubebe	
8	2026-10-31T11:24:13Z	AROA6GBMBODSEHTKAK244:user3801324=Lewis_Ubebe	{"queryExecutionIds":["1c4ae42e-f737-4e49-83c8-74543ff21c36","b3ae9fb2-fbc2-4531-9af6-3b5b44cc20fe","77c95e77-ef0a-47b3-a448-e98b56076eb4"]}
9	2026-10-31T11:24:13Z	AROA6GBMBODSEHTKAK244:user3801324=Lewis_Ubebe	{"bucketName":"athena-results-419325","Host":"athena-results-419325.s3.amazonaws.com","key":"Unsaved/2026/10/30/b3ae9fb2-fbc2-4531-9af6-3b5b44cc20f
10	2026-10-31T11:17:08Z		{"bucketName":"cloudtrail-logs-0cc0affea4f6e53fb","Host":"cloudtrail-logs-0cc0affea4f6e53fb.s3.us-east-1.amazonaws.com","x-amz-acl":"bucket-owner-full-contro



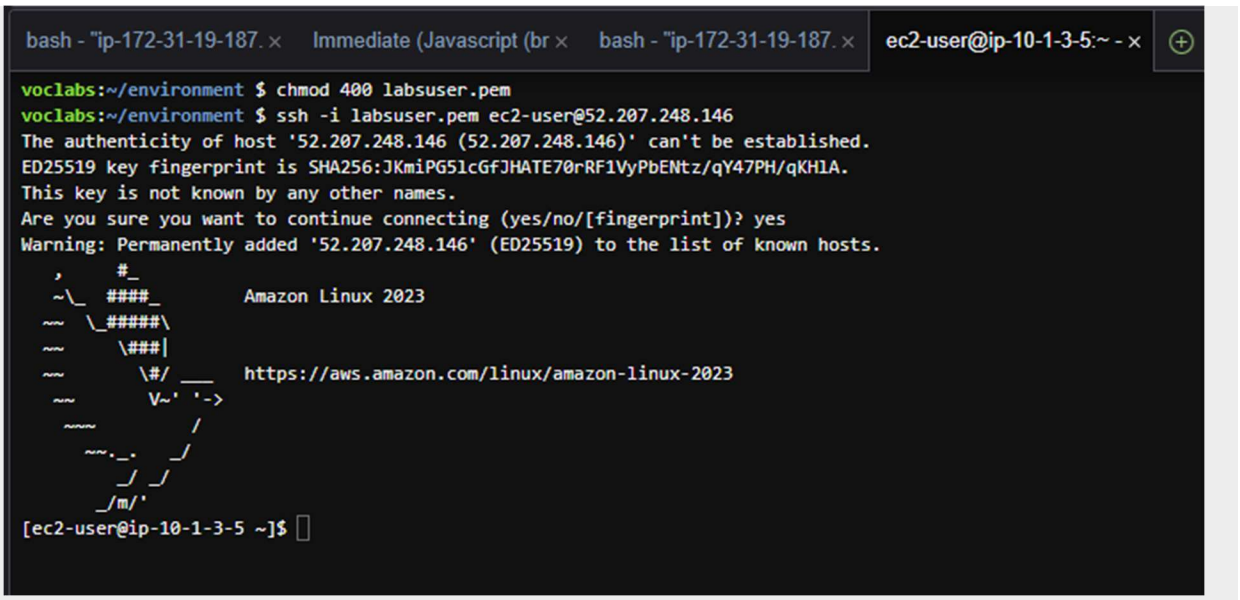


Template/configuration, CLI connection to EncryptedInstance and unauthorised username permission-denied evidence

```
sh-5.2$ sudo cat /opt/aws/amazon-cloudwatch-agent/bin/config.json
{
   "agent": {
      "metrics_collection_interval": 60,
      "run_as_user": "root"
   },
   "logs": {
      "logs_collected": {
            "files": {
               "collect_list": [
               {
                  "file_path": "/var/log/secure",
                  "log_group_name": "EncryptedInstanceSecureLogs",
                  "log_stream_name": "EncryptedInstanceSecureLogs-{instance_id}",
                  "retention_in_days": 180
               }
               ]
            }
      }
   },
   "metrics": {
      "aggregation_dimensions": [
         [
                              "InstanceId"
                     ]
               ],
               "append_dimensions": {
                     "AutoScalingGroupName": "${aws:AutoScalingGroupName}",
                     "ImageId": "${aws:ImageId}",
                     "InstanceId": "${aws:InstanceId}",
                     "InstanceType": "${aws:InstanceType}"
               },
               "metrics_collected": {
                     "collectd": {
                           "metrics_aggregation_interval": 60
                     },
                     "disk": {
                           "measurement": [
                                 "used_percent"
                           ],
                           "metrics_collection_interval": 60,
                           "resources": [
                                 "*"
                           ],
                           "ignore_file_system_types": [
                              "sysfs", "devtmpfs"
                           ]
                     },
                     "mem": {
                           "measurement": [
                                 "mem_used_percent"
                           ],
                           "metrics_collection_interval": 60
                     },
                     "statsd": {
                           "metrics_aggregation_interval": 60,
                           "metrics_collection_interval": 10,
                           "service_address": ":8125"
                     }
               }
```

```
bash - "ip-172-31-19-187. ×  Immediate (Javascript (br ×  bash - "ip-172-31-19-187. ×  ec2-user@ip-10-1-3-5:~ - ×

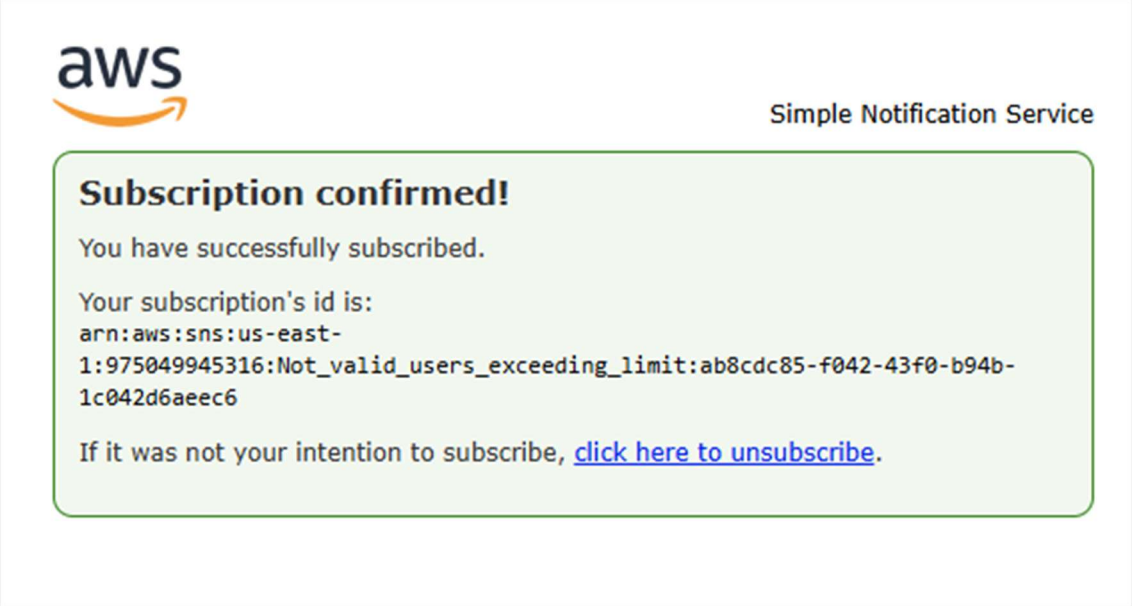
voclabs:~/environment $ chmod 400 labsuser.pem
voclabs:~/environment $ ssh -i labsuser.pem ec2-user@52.207.248.146
The authenticity of host '52.207.248.146 (52.207.248.146)' can't be established.
ED25519 key fingerprint is SHA256:JKmiPG51cGfJHATE70rRF1VyPbENtz/qY47PH/qKHlA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '52.207.248.146' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-1-3-5 ~]$
```


bash - "ip-172-31-19-187.
Immediate (Javascript (br
bash - "ip-172-31-19-187.
ec2-user@ip-10-1-3-5:~ -
voclabs:~/environment $ chmod 400 labsuser.pem
voclabs:~/environment $ ssh -i labsuser.pem ec2-user@52.207.248.146
The authenticity of host '52.207.248.146 (52.207.248.146)' can't be established.
ED25519 key fingerprint is SHA256:JKmiPG51cGfJHATE70rRF1VyPbENtz/qY47PH/qKHlA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '52.207.248.146' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-10-1-3-5 ~]$

R8 – Generate Alert and Change Management Notifications

A CloudWatch alarm was configured as part of the incident-detection solution so that security-relevant activity could trigger a notification. The alarm was connected to a notification subscription, providing a mechanism for the security team to be informed when the monitored condition was met.

CloudWatch Alarm subscription / notification configuration.
Best evidence: include the alarm name, metric/threshold and SNS subscription or notification destination in the screenshot.



This satisfies the requirement for automated security notification rather than relying on administrators to manually inspect logs. AWS Config remediation in R4 additionally provides controlled change management by detecting and correcting a resource that falls outside the required logging standard.

Cost and Security Observations

Amazon S3

The S3 security features can create additional cost. Versioning consumes additional storage because multiple versions are retained. Server access logging and S3 Inventory generate storage and request charges, while Athena queries can incur charges based on the volume of data scanned. In this lab the data volume was small, so the additional cost would be expected to remain low.

AWS KMS

AWS KMS can also create additional cost through the customer-managed key and KMS API operations such as encryption, decryption and data-key generation. Secrets Manager can add separate charges for secret storage and retrieval.

MONITORING AND COMPLIANCE

CloudTrail, CloudWatch and AWS Config can incur charges for data events, log ingestion and storage, metrics and alarms, resource configuration recording and compliance-rule evaluations. Actual cost depends on the amount of activity and data retained.

Conclusion

The completed solution applies defence in depth across data, identity, network and monitoring controls. Access to S3 data was restricted, network traffic was filtered at instance, subnet and firewall layers, sensitive data was encrypted using AWS KMS, and logging was implemented across S3, VPC, Network Firewall and CloudTrail. CloudWatch provided security monitoring and notification capability, while AWS Config identified and remediated noncompliant S3 logging configuration. The evidence presented above demonstrates coverage of solution requirements R1 through R8.